

ReToGCL: Reliability-Weighted Topology-Conditioned Graph Contrastive Learning for Medical Graph Classification

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Abstract

Medical graph classification must learn from heterogeneous graphs whose node attributes and local structures are not equally trustworthy. Standard graph contrastive learning typically treats two stochastic augmentations as equally valid positives and projects every graph into a single unconstrained geometry. These assumptions are fragile for brain-image patch graphs and molecular graphs: attribute corruption can erase clinically relevant signals, subgraph sampling can destabilize local motifs, and class imbalance can make high accuracy co-exist with poor minority-class structure. We introduce REToGCL, an end-to-end graph contrastive model that combines three ideas: reliable complementary augmentation, sample-view reliability learning, and topology-conditioned projection. An attribute-masked view and a structure-preserving sampled-subgraph view are encoded by a shared three-layer Graph Isomorphism Network. Node-, edge-, local-closure-, graph-, and depth-scale representations provide agreement, stability, and diversity evidence. Continuous reliability weights fuse rather than discard uncertain observations. A topology-conditioned bottleneck then organizes the fused representation before balanced contrastive learning with in-batch and hard negatives. After 40 self-supervised epochs, frozen representations are evaluated by a 200-epoch linear probe under stratified five-fold cross-validation. Across ADHD-200, BZR, COX2, AIDS, and the 12-task Tox21 benchmark, REToGCL obtains accuracies of 66.26, 83.93, 82.01, 99.40, and 92.71%, respectively. Its NMI/ARI/Macro-F1 are 5.88/9.90/62.86 on ADHD-200, 24.22/37.03/73.10 on BZR, 16.67/29.99/70.08 on COX2, 93.69/97.24/99.05 on AIDS, and 11.44/20.79/60.70 on Tox21. Under one unified project implementation, it strictly exceeds all four five-fold means of 34/35, 34/35, 33/35, 30/35, and 28/35 comparators, respectively. These are descriptive mean comparisons, not claims of statistical significance.

1 Introduction

Graphs provide a common language for biomedical systems. A brain image can be represented as a spatial graph of anatomical patches, while a molecule can be represented as atoms connected by chemical bonds. The

resulting graph-classification problems nevertheless differ sharply in graph size, feature semantics, class balance, and label completeness. ADHD-200 contains relatively large, fixed-lattice brain graphs; BZR and COX2 contain small and imbalanced molecular datasets; AIDS contains a pronounced class skew; and Tox21 combines twelve partially observed toxicity endpoints. A representation learner intended for this setting must therefore be robust to both perturbation noise and heterogeneous topology.

Graph contrastive learning (GCL) is attractive because it can pretrain an encoder without diagnostic or activity labels [CITATION NEEDED]. Most GCL pipelines maximize agreement between two augmented instances of the same graph and use other graphs as negatives. The critical but often implicit assumption is that both augmentations preserve task-relevant information. In medical graphs, that assumption is unreliable. Attribute masking may suppress a localized imaging or chemical signal; subgraph sampling may remove an important motif; and the same corruption strength may be benign for one graph but destructive for another. Uniformly weighting the two views transfers these errors into the contrastive target.

A second difficulty arises after encoding. Ordinary projection heads map all samples into one contrastive geometry. This can place same-class graphs far apart, move boundary samples toward a different class, or obscure a real within-class manifold with high-dimensional noise. These effects are especially consequential under imbalance: accuracy can remain high when a model predominantly predicts the majority class, while Macro-F1, normalized mutual information (NMI), and adjusted Rand index (ARI) expose weak class-balanced separation.

We address these issues with REToGCL, a reliability-weighted and topology-conditioned GCL framework. The model constructs complementary attribute-masked and sampled-subgraph views, encodes both using a shared three-layer GIN, and collects evidence at node, edge, local-closure, global-graph, and depth scales. It assigns a continuous reliability distribution to every sample-view pair. Thus, difficult graphs remain in training, but unreliable evidence contributes less. The reliability-fused representation conditions semantic routing and is subsequently organized through a topology-

conditioned bottleneck whose structural agreement is regularized. Balanced contrastive learning combines in-batch negatives with representation-dependent hard negatives. The entire pretraining system is optimized jointly, after which a linear classifier evaluates the frozen graph representation.

Our contributions are threefold:

- We formulate reliable complementary augmentation for medical graphs, coupling attribute masking and structure-preserving subgraph sampling without assuming that every perturbed view is equally faithful.
- We introduce sample-specific multi-view reliability learning from agreement, cross-augmentation stability, and representation diversity, retaining hard samples through continuous weighting rather than binary rejection.
- We combine the reliability mechanism with topology-conditioned projection and balanced contrastive learning in one end-to-end model. Under a unified PyG adaptation, data pipeline, and five-fold protocol, the combination improves medical graph representations across five heterogeneous benchmarks, as jointly assessed by Accuracy, NMI, ARI, and Macro-F1.

The empirical statements in this manuscript are deliberately bounded. “Exceeds” means that a reported five-fold mean is strictly larger. It does not imply statistical significance, and no significance test is claimed.

2 Related Work

2.1 Graph representation learning in biomedicine

Message-passing graph neural networks aggregate neighborhood information to form node and graph representations [CITATION NEEDED]. The Graph Isomorphism Network (GIN) is a widely used expressive backbone for graph classification [CITATION NEEDED]. Biomedical applications include molecular property prediction, drug activity classification, toxicity screening, and graph-based analysis of neuroimaging data [CITATION NEEDED]. These applications often combine limited labels with substantial imbalance and structurally heterogeneous samples. Consequently, aggregate accuracy alone is insufficient; class-balanced and clustering-oriented measures are also needed.

2.2 Graph contrastive learning and augmentation

GCL learns augmentation-invariant representations by matching multiple views of the same graph [CITATION NEEDED]. Common transformations include attribute masking, edge dropping, node dropping, and subgraph sampling. A fixed augmentation policy can be brittle because invariance is task- and sample-dependent. Adaptive augmentation methods learn global or instance-dependent perturbations [CITATION NEEDED], but many still treat the generated views as equally reliable after they have been produced. RETOGCL instead evaluates evidence at multiple graph granularities and assigns continuous reliability

to each sample-view observation.

2.3 Reliability, imbalance, and hard negatives

Noisy views and false negatives can bias a contrastive objective. Reliability estimation, uncertainty weighting, positive refinement, and hard-negative construction have therefore been investigated in self-supervised learning [CITATION NEEDED]. Class imbalance adds a related failure mode: an embedding can support majority-class accuracy while failing to separate minority examples. BalanceGCL is the direct foundation used in this project and is therefore included explicitly in the main comparison [CITATION NEEDED]. Our extension preserves balanced positive/negative construction while adding reliable augmentation, multi-view reliability learning, and topology-conditioned projection.

2.4 Topology-aware graph learning

Graph topology has been incorporated through motif statistics, higher-order structures, pooling, kernels, topological descriptors, and geometry-aware projection [CITATION NEEDED]. Local closure patterns are particularly useful because they distinguish tree-like neighborhoods from locally redundant or cyclic structures. However, forcing distinct topological regimes into one projection may distort neighborhood relations. RETOGCL uses local-closure and depth-scale evidence to condition the projection and explicitly discourages structural distortion.

The comparison pool contains BalanceGCL, HistoGraph, UniImb, DiffLift, MaxCutPool, CellCLAT, DualPrism, KAGNN, MultiNet, HSPGKN, DGDA, RSSN, TopoFormer, and 22 additional methods. Their bibliographic metadata must be verified before submission; all unverified citations are intentionally marked [CITATION NEEDED]. Importantly, the numerical values reported below are *not* copied from the corresponding papers. They are results of method-specific PyG adaptations run in the same project data pipeline and five-fold evaluation framework.

3 Problem Definition

Let a biomedical graph be $G_i = (V_i, E_i, X_i)$, where V_i is the node set, $E_i \subseteq V_i \times V_i$ is the undirected edge set, and $X_i \in \mathbb{R}^{|V_i| \times d_x}$ is the node-feature matrix. A dataset is $\mathcal{D} = \{(G_i, y_i)\}_{i=1}^N$, with a graph label $y_i \in \{1, \dots, C\}$. During self-supervised pretraining, labels are not used. The learner estimates an encoder f_θ such that a linear classifier trained on frozen $f_\theta(G)$ separates the downstream classes.

For Tox21, $y_{it} \in \{0, 1, \emptyset\}$ denotes the label for endpoint $t \in \{1, \dots, 12\}$. A missing value $\emptyset$ is excluded only from the corresponding endpoint. Each endpoint is split independently with stratification; its four metrics are computed separately, and the final Tox21 result is the unweighted macro-average over endpoints.

The representation problem is to learn a map

$$f_\theta : \mathcal{G} \rightarrow \mathbb{R}^{d_g} \quad (1)$$

that is invariant to semantics-preserving perturbations, sensitive to class-relevant structure, robust to unreliable views, and linearly useful after pretraining. No downstream label enters augmentation, reliability estimation, or contrastive pretraining.

4 Method

4.1 Overview

Figure 1 summarizes RETOGCL. For each input graph, two complementary perturbations are generated. A shared GIN encodes both views and exposes its layer history. Multi-granularity readouts produce node, edge, local-closure, and graph summaries as well as depth-scale summaries. Cross-view evidence yields a continuous reliability distribution. The fused representation conditions the downstream semantic organization, and a topology-conditioned bottleneck maps it to contrastive space. In-batch and hard negatives enter a balanced objective. All differentiable components are optimized jointly.

4.2 Reliable complementary augmentation

For each G_i , we generate

$$G_i^{(x)} = \mathcal{A}_x(G_i; p_x), \quad G_i^{(s)} = \mathcal{A}_s(G_i; p_s), \quad (2)$$

where $\mathcal{A}_x$ sets selected node-feature vectors to zero and $\mathcal{A}_s$ retains a density-favored sampled subset while masking removed nodes and incident edges. Both transformations preserve graph identity and enough original content to form a positive pair. Their complementarity is deliberate: the first probes attribute robustness while retaining topology; the second probes structural robustness while retaining the features of selected nodes. In the evaluated configuration, the augmentation budget is 0.10 and the minimum retained-content ratio is 0.60. These values are fixed across the reported runs.

We use the word *reliable* to describe the complete mechanism, not to assert that every generated view is clean. The generator exposes possibly noisy attributes and unstable substructures, after which the reliability module estimates how much each observation should influence training.

4.3 Shared GIN encoder and multi-layer fusion

Both views are processed by one parameter-shared three-layer GIN. With $h_v^{(0)} = x_v$, layer ℓ computes

$$h_v^{(\ell)} = \text{ReLU} \left(\text{BN} \left[\text{MLP}_\ell \left((1 + \epsilon_\ell) h_v^{(\ell-1)} + \sum_{u \in \mathcal{N}(v)} h_u^{(\ell-1)} \right) \right] \right). \quad (3)$$

The hidden dimension is 32. We retain $H^{(0)}, \dots, H^{(3)}$ and concatenate sum-pooled layer summaries,

$$g_i = \big\|_{\ell=0}^3 \text{Pool} \left(H_i^{(\ell)} \right), \quad (4)$$

which prevents the deepest layer from being the sole source of graph information.

4.4 Multi-granularity evidence

Let $\bar{h}_v$ denote the concatenated node history. Four graph-level views are constructed:

$$n_i = \text{MeanPool} \{ \bar{h}_v : v \in V_i \}, \quad (5)$$

$$e_i = \text{MeanPool} \{ (\bar{h}_u + \bar{h}_v) / 2 : (u, v) \in E_i \}, \quad (6)$$

$$c_i = \text{MeanPool} \left\{ \frac{|\bar{h}_v - \mu_{\mathcal{N}(v)}|}{\sqrt{\deg(v)}} : v \in V_i \right\}, \quad (7)$$

$$g_i = \text{SumPool} \{ \bar{h}_v : v \in V_i \}. \quad (8)$$

Here c_i is a differentiable local-closure residual: low residual indicates locally redundant or smoothly closed neighborhoods, whereas high residual exposes unstable or boundary-like structure. Each summary is linearly transformed and ℓ_2 -normalized. In addition, the first, middle, and last GIN states are independently pooled as three depth-scale summaries. Thus, topology and semantics are represented at both granularity and depth axes.

4.5 Sample-view reliability learning

For granularity $j \in \{n, e, c, g\}$ and the two augmentations $a \in \{x, s\}$, let $v_{ij}^{(a)}$ be the normalized representation. We calculate

$$A_{ij} = \cos \left(v_{ij}^{(x)}, v_{ij}^{(s)} \right), \quad (9)$$

$$S_{ij} = \exp \left[-d_v^{-1} \left\| v_{ij}^{(x)} - v_{ij}^{(s)} \right\|_2^2 \right], \quad (10)$$

$$D_{ij} = \sqrt{\text{Var}_k \left(v_{ij,k}^{(x)} \right)} + \epsilon. \quad (11)$$

A measures directional agreement, S measures perturbation stability, and D prevents reliability from collapsing onto an uninformative constant representation. An evidence network produces

$$q_{ij} = \rho_\omega([A_{ij}, S_{ij}, D_{ij}]), \quad w_{ij} = \frac{\exp(q_{ij})}{\sum_r \exp(q_{ir})}. \quad (12)$$

The reliability-conditioned representations are

$$z_i^{(a)} = \sum_j w_{ij} v_{ij}^{(a)}. \quad (13)$$

The reliability floor is 0.05 and topology contributes one half of the evidence mixture in the evaluated configuration. Because w_{ij} is continuous and normalized, no sample is removed. The fused representation conditions semantic routing, positive refinement, and hard-negative construction downstream.

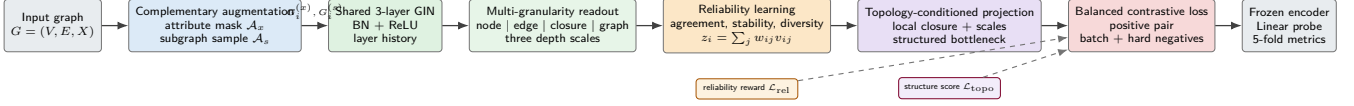


Figure 1: End-to-end REToGCL framework. The two augmented graphs share every encoder parameter. Continuous reliability weighting retains uncertain samples, while topology-conditioned projection organizes the representation before balanced contrastive learning. The linear probe is trained only after self-supervised pretraining.

We maximize a reliability reward

$$\mathcal{L}_{\text{rel}} = \frac{1}{B} \sum_{i=1}^B \sum_j w_{ij} A_{ij} + \eta H \left(\frac{1}{B} \sum_i w_i \right), \quad (14)$$

where the entropy term discourages batch-level collapse onto one granularity. This term is a reward (larger is better), which explains its negative sign in the minimized total objective below.

4.6 Topology-conditioned representation projection

Let t_i concatenate the closure view c_i and the three depth-scale summaries. A topology router assigns the graph to a soft mixture of K structural strata,

$$\pi_i = \text{softmax}(R_\psi(t_i)), \quad u_i = \sum_{k=1}^K \pi_{ik} P_k(z_i), \quad (15)$$

where P_k shares a common bottleneck but permits stratum-dependent modulation. The evaluated configuration uses $K = 16$, a bridge weight of 0.5, and a 32-dimensional intermediate bottleneck. A learnable projection head maps u_i to the normalized contrastive representation p_i .

To preserve topology during organization, define pairwise cosine-distance matrices Δ^{topo} from t and Δ^{proj} from p . We use the topology-preservation score

$$\mathcal{L}_{\text{topo}} = -\frac{1}{B^2} \|\Delta^{\text{proj}} - \text{sg}(\Delta^{\text{topo}})\|_F^2, \quad (16)$$

where sg stops gradients through the evidence target. Since the score is the negative distortion, maximizing it preserves pairwise structure. Conditioning before projection is intended to avoid forcing distinct topology types into an undifferentiated geometry and to reduce the effect of high-dimensional noise on local class structure.

4.7 Balanced graph contrastive learning

For graph i , the two reliable augmented representations form a positive pair. Other graphs in the minibatch provide negatives, and semantic alternatives generated from non-selected routing prototypes provide hard negatives h_{ik}^- . With cosine similarity $s(\cdot, \cdot)$ and temperature τ , one direction of the loss is

$$\begin{aligned} \ell_i^{x \rightarrow s} &= -\log \frac{\exp(s(p_i^{(x)}, p_i^{(s)})/\tau)}{Z_i}, \\ Z_i &= \sum_{r=1}^B \exp(s(p_i^{(x)}, p_r^{(s)})/\tau) \\ &\quad + \sum_k \exp(s(p_i^{(x)}, h_{ik}^-)/\tau). \end{aligned} \quad (17)$$

The contrastive term averages both directions:

$$\mathcal{L}_{\text{con}} = \frac{1}{2B} \sum_i (\ell_i^{x \rightarrow s} + \ell_i^{s \rightarrow x}). \quad (18)$$

The total minimized objective is

$$\mathcal{L} = \mathcal{L}_{\text{con}} - \lambda_{\text{clean}} \mathcal{L}_{\text{rel}} - \lambda_{\text{topo}} \mathcal{L}_{\text{topo}}. \quad (19)$$

Both auxiliary quantities in Eq. (19) are written as rewards. Therefore, the sign convention is equivalent to adding non-negative reliability and topology-distortion penalties. The evaluated implementation uses an auxiliary coefficient of 0.25 per mechanism.

4.8 Overall formulation and optimization

Combining Eqs. (3)–(19), the end-to-end map is

$$\begin{aligned} G_i &\xrightarrow{\mathcal{A}_x, \mathcal{A}_s} (G_i^{(x)}, G_i^{(s)}) \xrightarrow{f_\theta} \{v_{ij}^{(a)}, t_i^{(a)}\}_{j,a} \\ &\xrightarrow{\rho_\omega} \{z_i^{(a)}\}_a \xrightarrow{P_\psi(\cdot|t_i)} \{p_i^{(a)}\}_a \xrightarrow{\mathcal{L}} (\theta, \omega, \psi)^*. \end{aligned} \quad (20)$$

The downstream probe is deliberately excluded from self-supervised optimization.

5 Experimental Setup

5.1 Datasets

Table 1 summarizes the five medical benchmarks. ADHD-200 is a binary structural-MRI graph-classification task. Each subject is represented by a fixed $6 \times 6 \times 6$ patch lattice with 216 nodes and 540 undirected edges. Node features contain patch intensity mean, standard deviation, nonzero ratio, and normalized three-dimensional coordinates. The 655 labeled subjects comprise 405 healthy controls and 250 participants with ADHD.

BZR and COX2 are binary molecular-activity datasets. AIDS predicts antiviral activity. Tox21 contains twelve official binary toxicity endpoints. We do not impute missing Tox21 labels; a compound is excluded only for an endpoint whose label is missing. Dataset sources and canonical preprocessing descriptions require bibliographic completion [CITATION NEEDED].

5.2 Protocol and implementation

Self-supervised pretraining lasts 40 epochs. The frozen graph encoder is evaluated by a 200-epoch linear probe. We use stratified five-fold cross-validation with random seed 42. The encoder has three GIN layers, hidden dimension 32, ReLU activations, batch normalization, and

Table 1: Dataset statistics used in the unified project pipeline. Edge counts are undirected. Tox21 class distributions differ by endpoint and are therefore not summarized as one binary count.

Dataset	Graphs	Class distribution	Avg. nodes	Avg. edges	Feature dim.	Task
ADHD-200	655	405 / 250	216.00	540.00	6	ADHD vs. healthy control
BZR	405	319 / 86	35.75	38.36	53	Benzodiazepine-receptor activity
COX2	467	365 / 102	41.22	43.45	35	COX-2 inhibitor activity
AIDS	2,000	400 / 1,600	15.69	16.19	38	HIV/AIDS antiviral activity
Tox21	7,823	endpoint-specific	18.57	19.29	9	12 binary toxicity endpoints

Algorithm 1 End-to-end pretraining and linear-probe evaluation

Require: Graph dataset $\mathcal{D}$; encoder f_θ ; reliability module ρ_ω ; topology-conditioned projector P_ψ

- 1: **for** $e = 1$ to 40 **do**
- 2: **for** minibatch $\mathcal{B} \subset \mathcal{D}$ of size 128 **do**
- 3: **for** all $G_i \in \mathcal{B}$ **do**
- 4: $G_i^{(x)} \leftarrow \mathcal{A}_x(G_i)$; $G_i^{(s)} \leftarrow \mathcal{A}_s(G_i)$
- 5: Encode both views with the shared three-layer GIN
- 6: Read out node, edge, closure, graph, and depth-scale representations
- 7: Compute (A_{ij}, S_{ij}, D_{ij}) , reliability weights w_{ij} , and fused $z_i^{(x)}, z_i^{(s)}$
- 8: Compute topology evidence t_i and projected $p_i^{(x)}, p_i^{(s)}$
- 9: **end for**
- 10: Build in-batch and routing-based hard negatives
- 11: Evaluate $\mathcal{L}_{\text{con}}$, $\mathcal{L}_{\text{rel}}$, and $\mathcal{L}_{\text{topo}}$
- 12: Update (θ, ω, ψ) with Adam on Eq. (19)
- 13: **end for**
- 14: **end for**
- 15: Freeze f_{θ^*} and extract graph representations
- 16: **for** each stratified fold $k = 1, \dots, 5$ **do**
- 17: Train a linear classifier for 200 epochs on the fold-training representations
- 18: Evaluate Accuracy, NMI, ARI, and Macro-F1 on the held-out fold
- 19: **end for**
- 20: Report the mean and standard deviation across five folds

global pooling. The batch size is 128. Adam uses learning rate 10^{-3} and weight decay 10^{-5} . Every result is reported as the five-fold mean and standard deviation in percentage points.

The protocol is representation evaluation: self-supervised learning uses graph inputs without labels, and each linear probe accesses labels only in its training fold. For Tox21, the split and probe are constructed independently per endpoint, then the endpoint results are macro-averaged. Future work should additionally report a strictly inductive variant that pretrains separately inside each outer fold (Section 10).

The compared methods share the same project-level PyG interface, processed datasets, split generation, seed, and metric implementation. Their method-appropriate training mechanisms are retained. Consequently, the values in Table 2 are controlled project reproductions,

not numbers directly reported by the cited papers. The archived benchmark contains 35 comparators; the main table shows the 13 requested representative methods and the proposed model, while rank and dominance counts use all 35.

5.3 Metrics

Accuracy measures the fraction of correct predictions. Macro-F1 computes F1 independently per class and averages the class values, making it more sensitive to minority-class behavior. NMI measures the mutual dependence between predicted and true partitions after normalization. ARI corrects pairwise partition agreement for chance. For NMI and ARI, hard linear-probe predictions are treated as cluster assignments. Reporting all four prevents majority-class accuracy from being mistaken for uniformly good representation structure.

6 Main Results

6.1 Comparison with unified PyG adaptations

Table 2 reports all supplied main-table methods. The proposed row is shaded; this shading denotes the proposed method, not significance. The full 35-method archive is used for ranks in Table 3.

Table 2: Main results under the unified project adaptation and five-fold protocol (mean $\pm$ standard deviation, %). Values are project reproductions rather than original-paper reports. “Proposed” shading identifies REToGCL; it does not indicate statistical significance.

Dataset	Method	Accuracy	NMI	ARI	Macro-F1
ADHD-200	BalanceGCL	62.75 $\pm$ 6.04	3.78 $\pm$ 5.64	6.28 $\pm$ 8.20	58.61 $\pm$ 7.56
	HistoGraph	61.83 $\pm$ 0.00	0.00 $\pm$ 0.00	0.00 $\pm$ 0.00	38.21 $\pm$ 0.00
	UniImb	56.34 $\pm$ 10.29	0.10 $\pm$ 0.23	-0.32 $\pm$ 0.71	36.70 $\pm$ 5.24
	DiffLift	61.53 $\pm$ 0.68	0.01 $\pm$ 0.03	-0.10 $\pm$ 0.23	38.76 $\pm$ 1.23
	MaxCutPool	61.83 $\pm$ 0.00	0.00 $\pm$ 0.00	0.00 $\pm$ 0.00	38.21 $\pm$ 0.00
	CellCLAT	60.46 $\pm$ 3.56	1.39 $\pm$ 1.74	2.90 $\pm$ 3.20	54.31 $\pm$ 3.69
	DualPrism	63.36 $\pm$ 1.95	1.96 $\pm$ 2.16	3.65 $\pm$ 3.74	49.48 $\pm$ 10.66
	KAGNN	61.83 $\pm$ 0.00	0.00 $\pm$ 0.00	0.00 $\pm$ 0.00	38.21 $\pm$ 0.00
	MultiNet	61.83 $\pm$ 0.00	0.00 $\pm$ 0.00	0.00 $\pm$ 0.00	38.21 $\pm$ 0.00
	HSPGKN	62.75 $\pm$ 3.72	2.91 $\pm$ 2.35	4.21 $\pm$ 4.69	53.00 $\pm$ 9.24
	DGDA	62.29 $\pm$ 4.50	2.58 $\pm$ 2.69	4.09 $\pm$ 4.55	53.18 $\pm$ 8.82
	RSNN	61.83 $\pm$ 0.00	0.00 $\pm$ 0.00	0.00 $\pm$ 0.00	38.21 $\pm$ 0.00
	TopoFormer	61.83 $\pm$ 0.00	0.00 $\pm$ 0.00	0.00 $\pm$ 0.00	38.21 $\pm$ 0.00
	REToGCL	66.26 $\pm$ 4.23	5.88 $\pm$ 4.06	9.90 $\pm$ 5.83	62.86 $\pm$ 4.55
BZR	BalanceGCL	82.94 $\pm$ 6.90	21.18 $\pm$ 20.42	32.91 $\pm$ 23.02	70.32 $\pm$ 12.42
	HistoGraph	78.77 $\pm$ 0.42	0.00 $\pm$ 0.00	0.00 $\pm$ 0.00	44.06 $\pm$ 0.13
	UniImb	67.16 $\pm$ 25.82	0.00 $\pm$ 0.00	0.00 $\pm$ 0.00	38.70 $\pm$ 11.94
	DiffLift	78.77 $\pm$ 0.42	0.00 $\pm$ 0.00	0.00 $\pm$ 0.00	44.06 $\pm$ 0.13
	MaxCutPool	78.77 $\pm$ 0.42	0.00 $\pm$ 0.00	0.00 $\pm$ 0.00	44.06 $\pm$ 0.13
	CellCLAT	80.73 $\pm$ 4.47	12.15 $\pm$ 11.89	20.05 $\pm$ 15.95	61.97 $\pm$ 10.46
	DualPrism	82.97 $\pm$ 5.63	16.64 $\pm$ 20.65	25.20 $\pm$ 25.54	63.26 $\pm$ 16.16
	KAGNN	78.77 $\pm$ 0.42	0.00 $\pm$ 0.00	0.00 $\pm$ 0.00	44.06 $\pm$ 0.13
	MultiNet	78.77 $\pm$ 0.42	0.00 $\pm$ 0.00	0.00 $\pm$ 0.00	44.06 $\pm$ 0.13
	HSPGKN	78.77 $\pm$ 0.42	0.00 $\pm$ 0.00	0.00 $\pm$ 0.00	44.06 $\pm$ 0.13
	DGDA	77.06 $\pm$ 4.22	0.03 $\pm$ 0.06	-0.49 $\pm$ 1.09	44.82 $\pm$ 1.57
	RSNN	78.77 $\pm$ 0.42	0.00 $\pm$ 0.00	0.00 $\pm$ 0.00	44.06 $\pm$ 0.13
	TopoFormer	78.77 $\pm$ 0.42	0.00 $\pm$ 0.00	0.00 $\pm$ 0.00	44.06 $\pm$ 0.13
	REToGCL	83.93 $\pm$ 6.42	24.22 $\pm$ 18.52	37.03 $\pm$ 19.92	73.10 $\pm$ 9.65
COX2	BalanceGCL	83.08 $\pm$ 1.46	18.23 $\pm$ 4.68	31.66 $\pm$ 5.09	70.48 $\pm$ 3.16
	HistoGraph	78.16 $\pm$ 0.46	0.00 $\pm$ 0.00	0.00 $\pm$ 0.00	43.87 $\pm$ 0.14
	UniImb	78.16 $\pm$ 0.46	0.00 $\pm$ 0.00	0.00 $\pm$ 0.00	43.87 $\pm$ 0.14
	DiffLift	78.16 $\pm$ 0.46	0.00 $\pm$ 0.00	0.00 $\pm$ 0.00	43.87 $\pm$ 0.14
	MaxCutPool	78.16 $\pm$ 0.46	0.00 $\pm$ 0.00	0.00 $\pm$ 0.00	43.87 $\pm$ 0.14
	CellCLAT	76.88 $\pm$ 1.51	0.80 $\pm$ 1.02	2.93 $\pm$ 3.79	48.75 $\pm$ 4.08
	DualPrism	78.16 $\pm$ 0.46	0.00 $\pm$ 0.00	0.00 $\pm$ 0.00	43.87 $\pm$ 0.14
	KAGNN	78.16 $\pm$ 0.46	0.00 $\pm$ 0.00	0.00 $\pm$ 0.00	43.87 $\pm$ 0.14
	MultiNet	78.16 $\pm$ 0.46	0.00 $\pm$ 0.00	0.00 $\pm$ 0.00	43.87 $\pm$ 0.14
	HSPGKN	78.16 $\pm$ 0.46	0.00 $\pm$ 0.00	0.00 $\pm$ 0.00	43.87 $\pm$ 0.14
	DGDA	68.05 $\pm$ 22.42	0.10 $\pm$ 0.23	0.76 $\pm$ 1.71	40.61 $\pm$ 7.24
	RSNN	78.16 $\pm$ 0.46	0.00 $\pm$ 0.00	0.00 $\pm$ 0.00	43.87 $\pm$ 0.14
	TopoFormer	78.16 $\pm$ 0.46	0.00 $\pm$ 0.00	0.00 $\pm$ 0.00	43.87 $\pm$ 0.14
	REToGCL	82.01 $\pm$ 2.12	16.67 $\pm$ 5.57	29.99 $\pm$ 4.00	70.08 $\pm$ 1.30
AIDS	BalanceGCL	99.45 $\pm$ 0.45	94.20 $\pm$ 4.37	97.49 $\pm$ 2.04	99.14 $\pm$ 0.70
	HistoGraph	93.50 $\pm$ 1.08	56.47 $\pm$ 5.40	71.97 $\pm$ 3.92	89.77 $\pm$ 1.46
	UniImb	96.85 $\pm$ 2.97	77.68 $\pm$ 14.87	85.51 $\pm$ 13.92	94.52 $\pm$ 5.76
	DiffLift	79.75 $\pm$ 0.43	0.05 $\pm$ 0.09	-0.15 $\pm$ 0.20	44.60 $\pm$ 0.38
	MaxCutPool	99.30 $\pm$ 0.27	92.81 $\pm$ 2.45	96.79 $\pm$ 1.26	98.89 $\pm$ 0.44
	CellCLAT	99.30 $\pm$ 0.21	92.93 $\pm$ 1.71	96.78 $\pm$ 0.96	98.89 $\pm$ 0.34
	DualPrism	91.65 $\pm$ 2.00	48.64 $\pm$ 7.87	65.48 $\pm$ 7.18	87.34 $\pm$ 2.81
	KAGNN	92.70 $\pm$ 2.46	52.76 $\pm$ 11.90	69.17 $\pm$ 9.89	88.65 $\pm$ 4.00
	MultiNet	86.65 $\pm$ 5.05	27.23 $\pm$ 19.28	39.93 $\pm$ 25.34	72.42 $\pm$ 16.64
	HSPGKN	96.60 $\pm$ 0.80	72.62 $\pm$ 5.12	84.71 $\pm$ 3.63	94.55 $\pm$ 1.36
	DGDA	92.85 $\pm$ 1.58	52.26 $\pm$ 7.09	68.95 $\pm$ 6.77	88.41 $\pm$ 2.91
	RSNN	99.30 $\pm$ 0.60	93.37 $\pm$ 5.18	96.78 $\pm$ 2.75	98.88 $\pm$ 0.96
	TopoFormer	94.40 $\pm$ 1.01	61.03 $\pm$ 5.02	76.06 $\pm$ 4.04	91.48 $\pm$ 1.51
	REToGCL	99.40 $\pm$ 0.28	93.69 $\pm$ 2.40	97.24 $\pm$ 1.31	99.05 $\pm$ 0.46
Tox21	BalanceGCL	92.64 $\pm$ 4.40	9.41 $\pm$ 11.20	16.88 $\pm$ 16.70	58.38 $\pm$ 9.06
	HistoGraph	92.65 $\pm$ 4.65	9.39 $\pm$ 13.33	15.59 $\pm$ 20.24	57.22 $\pm$ 11.42
	UniImb	91.92 $\pm$ 5.36	1.15 $\pm$ 3.62	2.65 $\pm$ 7.77	49.87 $\pm$ 4.93
	DiffLift	92.25 $\pm$ 4.76	1.68 $\pm$ 4.77	3.40 $\pm$ 9.15	50.10 $\pm$ 5.50
	MaxCutPool	92.38 $\pm$ 4.80	4.63 $\pm$ 10.70	7.54 $\pm$ 16.92	52.15 $\pm$ 9.61
	CellCLAT	92.13 $\pm$ 4.77	2.09 $\pm$ 2.65	4.74 $\pm$ 5.33	51.34 $\pm$ 3.11
	DualPrism	92.66 $\pm$ 4.51	8.67 $\pm$ 12.19	14.87 $\pm$ 19.21	56.97 $\pm$ 10.80
	KAGNN	92.50 $\pm$ 4.66	5.78 $\pm$ 11.09	9.67 $\pm$ 18.12	53.49 $\pm$ 10.50
	MultiNet	92.34 $\pm$ 4.83	3.99 $\pm$ 10.07	6.41 $\pm$ 15.72	51.42 $\pm$ 8.93
	HSPGKN	92.38 $\pm$ 4.89	6.07 $\pm$ 12.05	10.05 $\pm$ 18.38	53.85 $\pm$ 10.39
	DGDA	92.54 $\pm$ 4.58	7.14 $\pm$ 11.29	11.81 $\pm$ 18.25	54.90 $\pm$ 10.42
	RSNN	92.67 $\pm$ 4.56	7.99 $\pm$ 12.33	12.94 $\pm$ 19.11	55.53 $\pm$ 11.09
	TopoFormer	92.43 $\pm$ 4.76	6.21 $\pm$ 10.70	10.95 $\pm$ 17.75	54.40 $\pm$ 10.27
	REToGCL	92.71 $\pm$ 4.37	11.44 $\pm$ 11.87	20.79 $\pm$ 16.62	60.70 $\pm$ 8.90

6.2 Ranks and strict-mean dominance

Table 3 uses all 35 archived comparators. On ADHD-200, RETOGCL ranks second for all four metrics and strictly exceeds all four means of 34 comparators. On BZR, NMI ranks first; the other metrics rank second, and 34 methods are jointly dominated. On COX2, each metric ranks third and the proposed model jointly exceeds 33 methods. AIDS accuracy reaches 99.40%; the corresponding ranks are fifth for Accuracy and NMI and sixth for ARI and Macro-F1. Tox21 shows the importance of looking beyond Accuracy: Accuracy ranks eighth, but ARI and Macro-F1 both rank first, while NMI ranks second. Across all five datasets simultaneously, 23 of the 35 methods have lower means on every one of the 20 dataset-metric combinations.

6.3 Dataset-level interpretation

ADHD-200. The 66.26 ± 4.23 Accuracy and 62.86 ± 4.55 Macro-F1 indicate that improvement is not confined to the majority control class. NMI and ARI remain modest in absolute terms (5.88 and 9.90), emphasizing that subject-level brain graphs remain difficult to partition cleanly. Their second-place ranks should be read as relative benchmark performance, not as evidence of a clinically separable embedding.

BZR. The model reaches 83.93 ± 6.42 Accuracy, 24.22 ± 18.52 NMI, 37.03 ± 19.92 ARI, and 73.10 ± 9.65 Macro-F1. NMI ranks first among all methods. The large NMI and ARI deviations reveal fold sensitivity in this small, imbalanced dataset; the mean advantage is descriptive and should not be interpreted as significant.

COX2. The four means are 82.01, 16.67, 29.99, and 70.08. Although BalanceGCL is higher in the selected table on each mean, RETOGCL exceeds 33 of 35 comparators on all four metrics. Its relatively small Macro-F1 standard deviation (1.30) suggests consistent folds in this run, but formal variance or paired significance comparisons were not conducted.

AIDS. Accuracy reaches 99.40 ± 0.28 , accompanied by 99.05 ± 0.46 Macro-F1, 93.69 ± 2.40 NMI, and 97.24 ± 1.31 ARI. The simultaneous strength of all four metrics argues against explaining the result solely through the majority class. Nevertheless, several comparators have higher means, and molecular scaffold or out-of-distribution splits were not evaluated.

Tox21. The 92.71 ± 4.37 Accuracy is less discriminative because of endpoint-specific imbalance. The first-place ARI (20.79 ± 16.62) and Macro-F1 (60.70 ± 8.90), together with second-place NMI (11.44 ± 11.87), provide more relevant evidence of class-balanced structure. Large standard deviations reflect heterogeneity across folds and the macro-aggregation of twelve tasks.

6.4 Improvement over the direct foundation

Relative to BalanceGCL, RETOGCL increases all four means on ADHD-200, BZR, and Tox21. For example, Tox21 improves from 58.38 to 60.70 Macro-F1 and from 16.88 to 20.79 ARI, while Accuracy changes from 92.64 to 92.71. This pattern is consistent with the intended effect under imbalance: representation structure can improve more than majority-dominated Accuracy. BalanceGCL remains higher on all four COX2 means and on all four AIDS means. Thus, the proposed additions are not uniformly superior to their foundation; they trade a small loss on those datasets for broad gains elsewhere. No statistical test has been performed to determine whether any pairwise difference is reliable beyond the observed folds.

7 Ablation Study Plan

No ablation experiment has yet been run for the five-dataset study. Table 4 is a planned design, not a result table. The primary question is whether the three contributions are complementary rather than whether any single component explains all gains. Every variant will use the same seed schedule, folds, pretraining budget, probe budget, and evaluation metrics as the main experiment.

The analysis will report per-dataset differences in all four metrics, not only Accuracy. Reliability-specific diagnostics should include the distribution and entropy of w_{ij} , the correlation between weights and cross-view perturbation magnitude, and minority/majority weight distributions. Topology diagnostics should compare pre/post-projection neighborhood overlap and pairwise-distance distortion. Because the same folds can be aligned across variants, paired confidence intervals or a suitable paired nonparametric analysis may be added after the experiments; no such result is presently claimed.

8 Parameter Sensitivity Plan

No sensitivity sweep has been conducted for the reported five-dataset results. We plan a one-factor-at-a-time screen followed by a limited joint search around stable regions. The default is marked in Table 5. Dataset splits and training budgets will remain fixed during screening, and model selection will not use the held-out test folds.

For each sweep, we will plot the four metrics with fold-level uncertainty, training stability, and peak memory. Particular attention will be paid to non-monotonic behavior: strong augmentation may improve invariance but destroy molecular motifs; a large reliability floor may suppress adaptivity; too many topology strata may overfit small datasets; and a narrow bottleneck may remove both noise and useful minority-class information. Joint $\lambda_{\text{clean}} \times \lambda_{\text{topo}}$ heatmaps are planned because the two rewards may interact.

Table 3: Ranks among REToGCL plus 35 comparators and the number of comparators whose four means are all strictly lower on each dataset. Ties are not counted as strict wins.

Dataset	Acc. rank	NMI rank	ARI rank	Macro-F1 rank	All-four strict wins
ADHD-200	2	2	2	2	34/35
BZR	2	1	2	2	34/35
COX2	3	3	3	3	33/35
AIDS	5	5	6	6	30/35
Tox21	8	2	1	1	28/35

Table 4: Planned ablations. All entries are unexecuted; no numerical outcome is implied. RA: reliable augmentation; MVR: multi-view reliability; TCP: topology-conditioned projection; HN: hard negatives.

Variant	RA	MVR	TCP	HN	Question	Status
Full REToGCL	✓	✓	✓	✓	Reference configuration	<i>Not run</i>
Uniform view weights	✓	–	✓	✓	Does sample-specific weighting matter?	<i>Not run</i>
Random/fixed augmentations	–	✓	✓	✓	Does reliable complementary augmentation matter?	<i>Not run</i>
Plain MLP projection	✓	✓	–	✓	Does topology conditioning improve geometry?	<i>Not run</i>
No topology score	✓	✓	partial	✓	Is explicit relation preservation necessary?	<i>Not run</i>
No hard negatives	✓	✓	✓	–	Do routed hard negatives add value?	<i>Not run</i>
Attribute-mask only	partial	✓	✓	✓	Is subgraph complementarity useful?	<i>Not run</i>
Subgraph-only	partial	✓	✓	✓	Is attribute complementarity useful?	<i>Not run</i>
No local-closure view	✓	✓	partial	✓	Is closure evidence necessary?	<i>Not run</i>
Last GIN layer only	✓	✓	partial	✓	Does multi-depth fusion help?	<i>Not run</i>
Binary sample filtering	✓	binary	✓	✓	Is continuous retention preferable to deletion?	<i>Not run</i>

9 Discussion

9.1 Why the three mechanisms are complementary

Reliable augmentation, multi-view reliability, and topology-conditioned projection act at different stages. Augmentation determines what invariances the encoder observes. Reliability learning decides how much each observation should influence the representation. Topology conditioning decides how the retained information is organized before contrast. A topology projector cannot recover semantics already deleted by an excessively destructive view; reliability weighting alone cannot repair a projection that collapses different structural regimes; and benign augmentations do not by themselves prevent hard negatives from distorting the contrastive geometry. Their end-to-end combination is therefore the central contribution.

9.2 Reading four metrics together

The results illustrate why one metric is insufficient. Tox21 Accuracy is high for many methods, yet NMI, ARI, and Macro-F1 vary substantially. Several baseline rows show near-majority-class behavior: Accuracy remains close to the class prior while NMI and ARI approach zero and Macro-F1 is low. In contrast, REToGCL improves Tox21 ARI and Macro-F1 ranks without producing an equally dramatic Accuracy change. AIDS is different: all four metrics are high, supporting both classification performance and partition agreement. ADHD-200 remains difficult in absolute NMI and ARI despite strong relative ranks.

9.3 Clinical and chemical interpretation

The mechanism is domain-general but not domain-agnostic. In ADHD-200, attribute masking challenges intensity statistics while the fixed spatial lattice is largely retained; subgraph sampling challenges spatial continuity. In molecules, attribute masking challenges atom descriptors and subgraph sampling challenges local chemical motifs. Reliability weighting can down-weight a damaged view, but it does not establish that the retained features are causal biomarkers or pharmacophores. Likewise, topology strata are learned representation devices, not clinically validated subtypes or chemical mechanism labels.

9.4 Reproducibility and comparison scope

Using one project implementation reduces differences caused by loaders, split generators, and metric code. It does not make all algorithms mathematically identical or remove all adaptation bias. Some methods are naturally self-supervised; others are supervised or require method-specific objectives. We therefore describe the table as a unified project benchmark rather than as a reproduction of claims in the original papers. The full archived outputs should accompany a submission, together with environment files, exact split indices, and method-specific adaptation notes.

10 Limitations

First, the reported evidence consists of five-fold means and standard deviations from one seed. Fold variation can be large, especially for BZR and Tox21, and no statistical significance test has been performed. Second,

Table 5: Planned parameter-sensitivity grid. The table specifies future runs only.

Parameter	Planned values	Current/default	Status
Attribute/subgraph budget	{0.05, 0.10, 0.20, 0.30, 0.40}	0.10	<i>Not run</i>
Minimum retained ratio	{0.40, 0.50, 0.60, 0.70, 0.80}	0.60	<i>Not run</i>
Reliability floor	{0, 0.01, 0.05, 0.10, 0.20}	0.05	<i>Not run</i>
Topology evidence ratio	{0, 0.25, 0.50, 0.75, 1.00}	0.50	<i>Not run</i>
λ_{clean}	{0, 0.05, 0.10, 0.25, 0.50, 1.00}	0.25	<i>Not run</i>
λ_{topo}	{0, 0.05, 0.10, 0.25, 0.50, 1.00}	0.25	<i>Not run</i>
Topology strata K	{4, 8, 16, 32}	16	<i>Not run</i>
Bottleneck dimension	{8, 16, 32, 64}	32	<i>Not run</i>
Temperature τ	{0.10, 0.20, 0.50, 1.00}	0.20	<i>Not run</i>
GIN depth	{2, 3, 4, 5}	3	<i>Not run</i>
Hidden dimension	{16, 32, 64, 128}	32	<i>Not run</i>
Batch size	{32, 64, 128, 256}	128	<i>Not run</i>

the ablation and sensitivity studies in Sections 7 and the following section are plans; they must be completed before making component-level causal claims. Third, self-supervised pretraining is performed without labels before fold-wise linear probing. Although probe labels are isolated by fold, a stricter inductive protocol would repeat pretraining using only each outer training fold. Fourth, random stratified molecular splits do not measure scaffold shift or prospective generalization. Fifth, Tox21 macro-aggregation may hide endpoint-specific failure, and missing labels are assumed ignorable after task-wise exclusion. Sixth, ADHD-200 uses one fixed patch-graph construction and does not establish robustness to acquisition site, preprocessing choice, atlas, or demographic confounding.

Seventh, local-closure residuals and topology strata are computational proxies rather than exact topological invariants. Their interpretability should not be overstated. Eighth, the 35 comparator values arise from project-specific PyG adaptations; adaptation quality may differ across methods, and the numbers must not be attributed to the original publications. Ninth, the paper presently contains citation placeholders because bibliographic metadata has not been independently verified. These placeholders must be replaced with checked primary sources before submission. Finally, high benchmark performance does not imply clinical readiness, biological mechanism discovery, calibrated risk, or safety in deployment.

11 Conclusion

We presented ReToGCL, an end-to-end contrastive framework for medical graph classification. Its core thesis is that reliable augmentation, multi-view reliability learning, and topology-conditioned projection should be optimized together. Complementary views expose attribute and structural instability; continuous reliability weights retain difficult graphs while reducing the influence of damaged observations; and topology-conditioned projection organizes the representation before balanced contrast with in-batch and hard negatives. Across ADHD-200, BZR, COX2, AIDS, and Tox21, the model provides competitive five-fold Accuracy, NMI, ARI, and Macro-F1, including first-ranked BZR NMI and first-ranked Tox21 ARI and Macro-F1 in the 35-method

project benchmark. These comparisons concern strict means only and do not establish statistical significance. The next required steps are the planned ablations, sensitivity sweeps, multi-seed uncertainty analysis, strictly inductive pretraining, and domain-specific external validation.

A Candidate Titles and Model Acronyms

1. **ReToGCL: Reliability-Weighted Topology-Conditioned Graph Contrastive Learning for Medical Graph Classification.** This is the title selected for the present draft; the acronym foregrounds reliability and topology.
2. **RAMP-GCL: Reliability-Aware Multi-View Projection for Contrastive Medical Graph Learning.** This option foregrounds the sample-view weighting and projection pipeline.
3. **TriMedGCL: Reliable Augmentation, Multi-View Reliability, and Topology-Conditioned Projection for Medical Graph Learning.** This option foregrounds the three-part contribution.

B Detailed Instructions for Redrawing the Framework Figure

The publication-ready framework should be redrawn as a horizontal vector diagram with an aspect ratio near 3.4:1 (for example, 2040×600 px or 17.8×5.2 cm). Use a white background, dark-gray 0.75–1.0 pt outlines, rounded rectangles, and a single sans-serif typeface. Keep the visual order strictly left to right:

1. **Input.** Draw one graph icon with 8–12 nodes and sparse undirected edges. Label it “Input medical graph $G = (V, E, X)$ ”. Optionally place a small brain-patch thumbnail and molecule thumbnail below it to indicate the two domains without implying that they are fused.
2. **Complementary augmentation.** Split the pipeline into two parallel lanes. The upper blue lane

is “Attribute masking” and should show the same graph with several node feature bars faded or crossed out. The lower blue lane is “Subgraph sampling” and should show a retained sampled subgraph with removed nodes in light gray. Add a brace labelled “reliable complementary views” and avoid calling either lane clean.

3. **Shared encoder.** Merge neither lane visually; instead, route both through two copies of the same green GIN block connected by a vertical equality/weight-sharing symbol. Each block contains three stacked mini-layers labelled GIN–BN–ReLU. Add “shared θ ” above the pair.
4. **Multi-granularity and multi-depth readout.** For each lane, draw four small tokens labelled Node, Edge, Closure, and Graph, plus three thinner bars labelled shallow, middle, deep. Join them into a common evidence panel. Use the closure icon as a triangle or locally closed neighborhood, but label it “local-closure proxy” rather than “persistent homology”.
5. **Reliability learning.** Use an orange panel. Place three gauges labelled Agreement A , Stability S , and Diversity D . Below them show a softmax weight vector such as $(0.18, 0.25, 0.41, 0.16)$ only as a schematic example and mark it “illustrative”. Show the equation $z_i^{(a)} = \sum_j w_{ij} v_{ij}^{(a)}$. Make line thickness or opacity from the four tokens to the fused vector proportional to the illustrative weights. Include a note “continuous weighting; no sample deletion”.
6. **Topology-conditioned projection.** Use a purple panel. At the top show closure evidence and three depth-scale bars feeding a small “topology router”. In the center draw several softly separated strata or curved sheets, then a narrow 32-dimensional bottleneck, and finally a compact contrastive plane. Label the relation-preservation arrow $\mathcal{L}_{\text{topo}}$. Avoid drawing ground-truth class labels inside this self-supervised block.
7. **Balanced contrastive objective.** Use a pale-red panel. Draw the two augmentations of one graph as a positive pair joined by a solid green line. Draw other minibatch graphs and routed hard negatives around the anchor with red repulsive arrows. Label the objective $\mathcal{L}_{\text{con}} - \lambda_{\text{clean}} \mathcal{L}_{\text{rel}} - \lambda_{\text{topo}} \mathcal{L}_{\text{topo}}$.
8. **Evaluation.** Use a neutral-gray panel separated by a dashed vertical line labelled “after pretraining”. Show a snowflake or lock on the encoder and a single linear layer leading to the four outputs Accuracy, NMI, ARI, and Macro-F1. Add “stratified 5-fold” beneath it.

Use blue for data perturbations, green for shared encoding, orange for reliability, purple for topology organization, red only for contrastive repulsion, and gray for evaluation. Solid arrows denote forward representations, dashed arrows denote auxiliary objectives, and a dotted arrow from topology evidence to the projector denotes conditioning. All equations should be typeset rather than embedded as raster text. The figure must remain

legible when printed at one-page width in grayscale; distinguish stages with labels and line styles in addition to color. Export as PDF or SVG with embedded fonts.

C Reproducibility Checklist

- Self-supervised epochs: 40; linear-probe epochs: 200.
- Five stratified folds; seed 42; all values reported as mean $\pm$ standard deviation.
- GIN depth 3; hidden dimension 32; ReLU; batch normalization; global pooling; shared view encoder.
- Batch size 128; Adam; learning rate 10^{-3} ; weight decay 10^{-5} .
- Tox21: 12 independently stratified binary endpoints; task-local exclusion of missing labels; no label imputation; macro-average over endpoints.
- Main-table baseline values: unified project PyG adaptations, not original-paper reports.
- Unperformed experiments: ablations, sensitivity sweeps, significance tests, multi-seed repeats, scaffold splits, and external validation.

Reference Completion Notice

No bibliographic record, author list, venue, year, or DOI has been inferred in this draft. Every unverified reference is shown literally as [CITATION NEEDED]. Before submission, each placeholder must be replaced with a verified primary-source entry, and dataset licenses and access statements must be added.

D Complete 18-Dataset SOTA Result Matrix

Table 6 reports all four metrics for the 35 comparator adaptations and ReToGCL on every graph-classification dataset currently supported by the project. This yields 648 completed method–dataset rows. The comparator numbers are outputs of the project’s unified PyG adaptation, data processing, and five-fold protocol; they must not be interpreted as values directly reported in the corresponding original papers. SIDER and Tox21 are macro-averages over 27 and 12 independently evaluated binary endpoints, respectively. ClinTox follows the project’s joint three-class construction and is not the official two-endpoint ROC–AUC protocol. All entries are percentages reported as mean $\pm$ standard deviation.

Pale-red rows expose a useful failure signature: NMI and ARI are both exactly 0.00 ± 0.00 , which is consistent with a constant predicted partition across folds. Such rows can retain apparently non-trivial accuracy on an imbalanced dataset by predicting only the majority class; hence Accuracy, NMI, ARI, and Macro-F1 must be interpreted jointly. Boldface marks only the largest completed five-fold mean in each dataset and metric, not statistical significance. The matrix is descriptive rather than a cross-protocol aggregate, and raw means should not be averaged indiscriminately across binary, multiclass, and multi-task settings.

Table 6: Complete unified-project results for 35 comparator adaptations and ReToGCL across 18 datasets (mean $\pm$ standard deviation, %). Bold marks the largest completed mean within each dataset and metric; it does not indicate significance. Pale-red rows have exactly zero NMI and ARI mean and standard deviation, a constant-partition signature. Purple rows denote ReToGCL.

Method	Accuracy	NMI	ARI	Macro-F1
NCI1				
BalanceGCL	74.48 $\pm$ 2.36	18.43 $\pm$ 3.71	24.05 $\pm$ 4.51	74.44 $\pm$ 2.34
HistoGraph	74.36 $\pm$ 3.31	18.88 $\pm$ 4.56	23.99 $\pm$ 6.09	74.18 $\pm$ 3.42
UniImb	65.01 $\pm$ 10.29	11.03 $\pm$ 9.01	12.34 $\pm$ 10.72	60.65 $\pm$ 16.77
DiffLift	64.35 $\pm$ 2.27	6.59 $\pm$ 1.64	8.30 $\pm$ 2.59	63.82 $\pm$ 2.75
KhanGCL	73.65 $\pm$ 2.58	17.31 $\pm$ 3.99	22.49 $\pm$ 4.78	73.59 $\pm$ 2.55
MaxCutPool	74.23 $\pm$ 2.91	18.42 $\pm$ 4.24	23.67 $\pm$ 5.28	74.13 $\pm$ 2.93
SimplicialMP	70.10 $\pm$ 3.20	13.48 $\pm$ 3.53	16.39 $\pm$ 5.15	69.56 $\pm$ 3.67
DEL	49.95 $\pm$ 0.03	0.00 $\pm$ 0.00	0.00 $\pm$ 0.00	33.31 $\pm$ 0.01
CellCLAT	66.52 $\pm$ 1.34	8.22 $\pm$ 1.40	10.87 $\pm$ 1.81	66.43 $\pm$ 1.33
pLSTM	71.59 $\pm$ 8.20	16.03 $\pm$ 8.23	20.70 $\pm$ 10.83	71.18 $\pm$ 8.98
AbstainGNN	72.82 $\pm$ 1.44	15.91 $\pm$ 1.97	20.80 $\pm$ 2.66	72.75 $\pm$ 1.49
EdgePrompt	73.65 $\pm$ 2.96	17.33 $\pm$ 4.00	22.56 $\pm$ 5.24	73.57 $\pm$ 3.02
DualPrism	70.56 $\pm$ 2.54	12.90 $\pm$ 3.16	17.01 $\pm$ 4.18	70.50 $\pm$ 2.56
InvGNN	62.41 $\pm$ 1.60	4.60 $\pm$ 1.17	6.13 $\pm$ 1.54	62.35 $\pm$ 1.57
GenVsExp	75.98 $\pm$ 3.23	21.20 $\pm$ 5.01	27.25 $\pm$ 6.39	75.90 $\pm$ 3.28
KAGNN	73.72 $\pm$ 0.94	17.07 $\pm$ 1.37	22.44 $\pm$ 1.81	73.69 $\pm$ 0.96
MultiNet	69.88 $\pm$ 2.28	12.00 $\pm$ 3.09	15.87 $\pm$ 3.85	69.83 $\pm$ 2.25
HSPGKN	73.16 $\pm$ 1.72	16.64 $\pm$ 2.84	21.46 $\pm$ 3.31	73.05 $\pm$ 1.70
DGDA	72.44 $\pm$ 3.53	15.87 $\pm$ 4.67	20.44 $\pm$ 6.15	72.29 $\pm$ 3.62
RSNN	71.07 $\pm$ 2.86	14.85 $\pm$ 2.92	17.93 $\pm$ 4.64	70.52 $\pm$ 3.44
TopoFormer	66.13 $\pm$ 2.03	8.27 $\pm$ 1.90	10.44 $\pm$ 2.65	65.74 $\pm$ 2.39
G2PM	75.67 $\pm$ 4.68	20.91 $\pm$ 7.55	26.97 $\pm$ 9.21	75.63 $\pm$ 4.67
Git-TaskTree	73.23 $\pm$ 3.52	17.63 $\pm$ 4.39	21.90 $\pm$ 6.28	72.87 $\pm$ 3.95
GNN+	77.84 $\pm$ 2.64	24.36 $\pm$ 5.20	31.13 $\pm$ 6.10	77.78 $\pm$ 2.65
Hourglass	76.79 $\pm$ 1.66	22.12 $\pm$ 2.62	28.71 $\pm$ 3.45	76.75 $\pm$ 1.70
HyperEnc	75.11 $\pm$ 2.19	20.52 $\pm$ 3.03	25.29 $\pm$ 4.38	74.84 $\pm$ 2.39
LEAP	74.82 $\pm$ 2.49	18.85 $\pm$ 4.03	24.74 $\pm$ 5.10	74.80 $\pm$ 2.51
MagEdgePool	72.04 $\pm$ 2.32	15.05 $\pm$ 2.85	19.51 $\pm$ 3.97	71.91 $\pm$ 2.44
NodeID	69.36 $\pm$ 4.20	11.92 $\pm$ 4.76	15.46 $\pm$ 6.19	69.21 $\pm$ 4.27
PRUNE	63.94 $\pm$ 3.12	6.20 $\pm$ 2.62	7.98 $\pm$ 3.60	63.59 $\pm$ 3.35
RankFeatures	74.53 $\pm$ 2.95	18.63 $\pm$ 4.26	24.25 $\pm$ 5.43	74.47 $\pm$ 2.97
RSPool	76.18 $\pm$ 1.38	20.99 $\pm$ 2.41	27.39 $\pm$ 2.91	76.16 $\pm$ 1.37
SPECTRE	77.03 $\pm$ 2.40	22.65 $\pm$ 4.06	29.33 $\pm$ 5.07	77.00 $\pm$ 2.40
TIF	73.77 $\pm$ 1.67	17.45 $\pm$ 2.41	22.60 $\pm$ 3.25	73.68 $\pm$ 1.72
TopER	74.09 $\pm$ 2.74	18.37 $\pm$ 4.34	23.36 $\pm$ 5.31	73.94 $\pm$ 2.77
ReToGCL	74.82 $\pm$ 0.80	18.78 $\pm$ 1.31	24.57 $\pm$ 1.58	74.78 $\pm$ 0.79
PROTEINS				
BalanceGCL	73.67 $\pm$ 3.35	16.13 $\pm$ 4.93	22.36 $\pm$ 6.14	72.77 $\pm$ 3.60
HistoGraph	70.35 $\pm$ 2.92	11.06 $\pm$ 3.79	16.12 $\pm$ 5.07	68.00 $\pm$ 3.63
UniImb	62.35 $\pm$ 33.11	35.81 $\pm$ 36.51	40.17 $\pm$ 39.80	58.16 $\pm$ 34.54
DiffLift	65.68 $\pm$ 2.13	5.52 $\pm$ 2.15	8.57 $\pm$ 2.78	59.58 $\pm$ 3.18
KhanGCL	73.40 $\pm$ 2.72	15.68 $\pm$ 4.03	21.72 $\pm$ 5.18	72.47 $\pm$ 2.79
MaxCutPool	75.38 $\pm$ 2.15	18.12 $\pm$ 3.59	25.28 $\pm$ 4.41	73.41 $\pm$ 2.39
SimplicialMP	75.47 $\pm$ 3.05	18.67 $\pm$ 5.10	25.61 $\pm$ 6.50	73.47 $\pm$ 4.06
DEL	59.57 $\pm$ 0.10	0.00 $\pm$ 0.00	0.00 $\pm$ 0.00	37.33 $\pm$ 0.04
CellCLAT	74.93 $\pm$ 2.85	17.45 $\pm$ 4.64	24.63 $\pm$ 5.84	73.59 $\pm$ 2.99
pLSTM	69.18 $\pm$ 2.68	9.48 $\pm$ 3.25	14.34 $\pm$ 4.35	67.29 $\pm$ 2.84
AbstainGNN	74.84 $\pm$ 1.88	17.17 $\pm$ 2.96	24.19 $\pm$ 3.91	72.95 $\pm$ 2.65
EdgePrompt	69.99 $\pm$ 2.45	10.36 $\pm$ 2.76	15.22 $\pm$ 4.29	66.33 $\pm$ 4.86
DualPrism	70.08 $\pm$ 2.16	10.44 $\pm$ 2.68	15.51 $\pm$ 3.40	67.42 $\pm$ 2.36
InvGNN	69.35 $\pm$ 5.79	11.19 $\pm$ 6.59	14.36 $\pm$ 8.61	61.06 $\pm$ 13.63
GenVsExp	70.44 $\pm$ 0.82	10.68 $\pm$ 1.03	15.99 $\pm$ 1.29	67.89 $\pm$ 0.77
KAGNN	71.43 $\pm$ 2.16	12.18 $\pm$ 2.95	17.73 $\pm$ 4.05	68.69 $\pm$ 3.37
MultiNet	71.61 $\pm$ 3.50	12.81 $\pm$ 4.79	18.50 $\pm$ 6.19	69.94 $\pm$ 3.94
HSPGKN	65.41 $\pm$ 4.23	5.64 $\pm$ 4.46	8.99 $\pm$ 6.22	60.89 $\pm$ 7.38
DGDA	68.64 $\pm$ 2.91	9.89 $\pm$ 3.61	13.70 $\pm$ 4.19	67.60 $\pm$ 3.23
RSNN	74.39 $\pm$ 3.67	16.94 $\pm$ 5.42	23.54 $\pm$ 7.04	72.08 $\pm$ 5.13
TopoFormer	72.14 $\pm$ 3.39	13.70 $\pm$ 4.58	19.15 $\pm$ 6.33	69.15 $\pm$ 4.69
G2PM	72.06 $\pm$ 2.15	13.35 $\pm$ 2.92	18.73 $\pm$ 4.21	68.85 $\pm$ 3.40
Git-TaskTree	71.60 $\pm$ 3.61	12.71 $\pm$ 4.90	18.40 $\pm$ 6.43	69.29 $\pm$ 5.28
GNN+	74.66 $\pm$ 2.41	18.25 $\pm$ 5.17	24.08 $\pm$ 4.87	73.74 $\pm$ 2.96
Hourglass	74.75 $\pm$ 3.06	17.14 $\pm$ 4.96	24.22 $\pm$ 6.18	72.96 $\pm$ 3.60
HyperEnc	72.86 $\pm$ 2.56	14.74 $\pm$ 3.97	20.47 $\pm$ 4.67	70.86 $\pm$ 2.93
LEAP	71.52 $\pm$ 2.99	12.90 $\pm$ 2.90	17.82 $\pm$ 5.45	67.63 $\pm$ 6.65
MagEdgePool	70.53 $\pm$ 2.77	11.12 $\pm$ 3.48	16.46 $\pm$ 4.61	68.51 $\pm$ 2.65
NodeID	67.39 $\pm$ 2.39	7.67 $\pm$ 2.87	11.17 $\pm$ 3.40	63.00 $\pm$ 2.93

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Table 6 — continued

Method	Accuracy	NMI	ARI	Macro-F1
PRUNE	70.80 $\pm$ 2.34	11.59 $\pm$ 2.85	16.83 $\pm$ 3.83	68.74 $\pm$ 2.00
RankFeatures	70.98 $\pm$ 3.63	12.15 $\pm$ 4.41	17.11 $\pm$ 6.10	67.47 $\pm$ 6.68
RSPool	72.23 $\pm$ 3.80	13.54 $\pm$ 5.02	19.58 $\pm$ 6.55	70.11 $\pm$ 4.84
SPECTRE	73.85 $\pm$ 2.12	15.70 $\pm$ 3.27	22.28 $\pm$ 4.15	71.87 $\pm$ 2.80
TIF	71.25 $\pm$ 1.86	11.86 $\pm$ 2.58	17.56 $\pm$ 3.32	69.36 $\pm$ 2.35
TopER	73.58 $\pm$ 2.28	15.37 $\pm$ 3.30	21.68 $\pm$ 4.45	71.09 $\pm$ 3.22
REToGCL	75.47 $\pm$ 2.75	18.78 $\pm$ 4.43	25.79 $\pm$ 5.59	74.63 $\pm$ 2.82
COLLAB				
BalanceGCL	70.72 $\pm$ 1.90	25.89 $\pm$ 2.64	28.38 $\pm$ 3.49	67.42 $\pm$ 1.86
HistoGraph	71.78 $\pm$ 2.73	28.63 $\pm$ 3.44	30.66 $\pm$ 4.71	68.72 $\pm$ 3.31
UniImb	85.18 $\pm$ 3.97	57.13 $\pm$ 6.39	58.47 $\pm$ 9.93	84.05 $\pm$ 4.24
DiffLift	65.30 $\pm$ 3.86	21.34 $\pm$ 4.57	17.61 $\pm$ 7.51	53.12 $\pm$ 10.73
KhanGCL	72.72 $\pm$ 1.98	27.92 $\pm$ 3.15	32.59 $\pm$ 3.88	69.68 $\pm$ 2.03
MaxCutPool	69.24 $\pm$ 1.96	25.89 $\pm$ 1.44	26.30 $\pm$ 2.84	64.65 $\pm$ 2.56
SimplicialMP	53.70 $\pm$ 2.44	2.44 $\pm$ 3.77	1.42 $\pm$ 1.95	27.66 $\pm$ 6.70
DEL	52.00 $\pm$ 0.00	0.00 $\pm$ 0.00	0.00 $\pm$ 0.00	22.81 $\pm$ 0.00
CellCLAT	56.18 $\pm$ 0.84	6.03 $\pm$ 1.18	2.85 $\pm$ 0.87	32.48 $\pm$ 1.31
pLSTM	80.38 $\pm$ 1.43	41.13 $\pm$ 3.49	49.41 $\pm$ 3.64	77.21 $\pm$ 1.55
AbstainGNN	71.12 $\pm$ 1.66	26.18 $\pm$ 2.79	28.83 $\pm$ 3.13	67.96 $\pm$ 2.13
EdgePrompt	72.48 $\pm$ 2.07	29.49 $\pm$ 3.10	32.10 $\pm$ 3.88	70.00 $\pm$ 1.90
DualPrism	70.26 $\pm$ 1.85	26.09 $\pm$ 2.79	27.75 $\pm$ 3.39	67.48 $\pm$ 3.04
InvGNN	62.20 $\pm$ 2.95	15.70 $\pm$ 5.40	10.56 $\pm$ 3.85	41.39 $\pm$ 4.01
GenVsExp	71.44 $\pm$ 1.67	29.90 $\pm$ 2.23	30.21 $\pm$ 2.91	67.70 $\pm$ 1.70
KAGNN	72.50 $\pm$ 1.56	28.60 $\pm$ 2.17	31.53 $\pm$ 2.86	69.67 $\pm$ 1.68
MultiNet	68.64 $\pm$ 1.52	25.48 $\pm$ 1.73	24.94 $\pm$ 1.83	63.15 $\pm$ 1.81
HSPGKN	69.88 $\pm$ 1.84	26.96 $\pm$ 3.11	27.33 $\pm$ 3.22	65.96 $\pm$ 0.79
DGDA	70.06 $\pm$ 1.34	27.89 $\pm$ 1.17	27.77 $\pm$ 1.85	65.78 $\pm$ 1.01
RSNN	74.96 $\pm$ 1.07	32.48 $\pm$ 1.72	36.80 $\pm$ 2.46	72.14 $\pm$ 1.99
TopoFormer	72.82 $\pm$ 1.85	30.86 $\pm$ 3.03	33.16 $\pm$ 3.51	71.01 $\pm$ 1.72
G2PM	70.20 $\pm$ 1.69	26.82 $\pm$ 2.24	27.85 $\pm$ 2.87	66.15 $\pm$ 1.77
Git-TaskTree	69.36 $\pm$ 2.35	25.85 $\pm$ 3.59	26.85 $\pm$ 3.93	65.97 $\pm$ 2.64
GNN+	70.56 $\pm$ 1.26	26.56 $\pm$ 2.37	28.47 $\pm$ 2.21	67.84 $\pm$ 1.81
Hourglass	69.56 $\pm$ 0.85	25.35 $\pm$ 1.82	26.85 $\pm$ 1.82	66.15 $\pm$ 0.74
HyperEnc	68.98 $\pm$ 1.62	25.29 $\pm$ 2.88	26.06 $\pm$ 2.59	65.52 $\pm$ 1.44
LEAP	72.78 $\pm$ 2.31	31.23 $\pm$ 3.18	33.02 $\pm$ 4.06	70.92 $\pm$ 2.11
MagEdgePool	52.00 $\pm$ 0.00	0.00 $\pm$ 0.00	0.00 $\pm$ 0.00	22.81 $\pm$ 0.00
NodeID	72.62 $\pm$ 0.30	29.29 $\pm$ 0.69	32.46 $\pm$ 0.40	69.71 $\pm$ 0.46
PRUNE	52.00 $\pm$ 0.00	0.00 $\pm$ 0.00	0.00 $\pm$ 0.00	22.81 $\pm$ 0.00
RankFeatures	72.58 $\pm$ 1.74	29.83 $\pm$ 3.14	32.35 $\pm$ 3.59	69.57 $\pm$ 1.83
RSPool	71.08 $\pm$ 2.85	27.35 $\pm$ 5.02	29.17 $\pm$ 5.31	67.61 $\pm$ 4.03
SPECTRE	72.84 $\pm$ 1.81	29.21 $\pm$ 2.10	32.76 $\pm$ 3.35	70.22 $\pm$ 1.61
TIF	70.62 $\pm$ 2.29	26.78 $\pm$ 3.75	28.55 $\pm$ 4.09	67.11 $\pm$ 2.32
TopER	72.08 $\pm$ 1.65	28.70 $\pm$ 2.24	31.39 $\pm$ 3.03	69.26 $\pm$ 2.72
REToGCL	73.78 $\pm$ 1.46	30.28 $\pm$ 2.23	34.31 $\pm$ 2.95	71.33 $\pm$ 1.40
MUTAG				
BalanceGCL	87.23 $\pm$ 4.73	47.15 $\pm$ 15.03	54.78 $\pm$ 14.94	86.11 $\pm$ 4.79
HistoGraph	66.50 $\pm$ 0.97	0.00 $\pm$ 0.00	0.00 $\pm$ 0.00	39.94 $\pm$ 0.35
UniImb	69.20 $\pm$ 6.68	6.85 $\pm$ 15.32	6.70 $\pm$ 14.99	46.53 $\pm$ 14.95
DiffLift	66.50 $\pm$ 0.97	0.00 $\pm$ 0.00	0.00 $\pm$ 0.00	39.94 $\pm$ 0.35
KhanGCL	85.62 $\pm$ 4.10	42.61 $\pm$ 12.94	49.64 $\pm$ 12.25	84.11 $\pm$ 4.02
MaxCutPool	69.13 $\pm$ 5.56	4.78 $\pm$ 10.68	6.33 $\pm$ 14.14	47.48 $\pm$ 16.72
SimplicialMP	71.88 $\pm$ 9.91	12.50 $\pm$ 23.88	14.33 $\pm$ 25.62	53.48 $\pm$ 20.58
DEL	33.50 $\pm$ 0.97	0.00 $\pm$ 0.00	0.00 $\pm$ 0.00	25.09 $\pm$ 0.55
CellCLAT	81.96 $\pm$ 9.47	31.98 $\pm$ 16.38	41.86 $\pm$ 20.91	80.07 $\pm$ 9.69
pLSTM	66.50 $\pm$ 0.97	0.00 $\pm$ 0.00	0.00 $\pm$ 0.00	39.94 $\pm$ 0.35
AbstainGNN	66.50 $\pm$ 0.97	0.00 $\pm$ 0.00	0.00 $\pm$ 0.00	39.94 $\pm$ 0.35
EdgePrompt	68.69 $\pm$ 8.75	9.88 $\pm$ 9.28	9.98 $\pm$ 12.57	55.29 $\pm$ 12.88
DualPrism	72.32 $\pm$ 6.96	12.89 $\pm$ 13.65	15.87 $\pm$ 15.83	61.52 $\pm$ 16.05
InvGNN	66.50 $\pm$ 0.97	0.00 $\pm$ 0.00	0.00 $\pm$ 0.00	39.94 $\pm$ 0.35
GenVsExp	67.55 $\pm$ 2.15	7.58 $\pm$ 13.80	4.55 $\pm$ 6.80	51.02 $\pm$ 15.17
KAGNN	68.63 $\pm$ 6.18	5.19 $\pm$ 7.37	7.25 $\pm$ 11.72	51.37 $\pm$ 14.40
MultiNet	65.97 $\pm$ 1.81	1.08 $\pm$ 1.49	0.55 $\pm$ 2.85	43.25 $\pm$ 7.84
HSPGKN	77.18 $\pm$ 6.94	25.91 $\pm$ 17.78	26.78 $\pm$ 17.06	68.67 $\pm$ 17.49
DGDA	72.82 $\pm$ 8.85	15.99 $\pm$ 24.14	15.06 $\pm$ 23.35	54.52 $\pm$ 21.02
RSNN	80.88 $\pm$ 5.91	30.45 $\pm$ 13.61	36.47 $\pm$ 14.80	77.26 $\pm$ 6.59
TopoFormer	76.10 $\pm$ 7.79	20.39 $\pm$ 15.91	25.60 $\pm$ 17.13	70.76 $\pm$ 10.15
G2PM	79.33 $\pm$ 8.67	28.86 $\pm$ 18.81	33.15 $\pm$ 21.94	72.42 $\pm$ 18.88
Git-TaskTree	70.18 $\pm$ 7.89	6.66 $\pm$ 14.89	8.95 $\pm$ 20.00	48.35 $\pm$ 18.66
GNN+	77.20 $\pm$ 7.40	22.77 $\pm$ 12.93	27.89 $\pm$ 16.67	72.17 $\pm$ 8.73
Hourglass	70.28 $\pm$ 9.09	9.68 $\pm$ 21.14	10.48 $\pm$ 22.19	49.70 $\pm$ 18.51
HyperEnc	69.66 $\pm$ 6.72	5.41 $\pm$ 12.09	7.55 $\pm$ 16.88	47.83 $\pm$ 17.50
LEAP	68.12 $\pm$ 4.29	2.97 $\pm$ 6.64	4.04 $\pm$ 9.04	45.00 $\pm$ 11.54
MagEdgePool	68.12 $\pm$ 4.29	3.83 $\pm$ 6.25	5.95 $\pm$ 8.33	51.45 $\pm$ 11.24
NodeID	72.90 $\pm$ 7.18	15.16 $\pm$ 18.81	16.58 $\pm$ 16.86	59.78 $\pm$ 19.03
PRUNE	66.50 $\pm$ 0.97	0.14 $\pm$ 0.31	0.33 $\pm$ 0.73	41.20 $\pm$ 2.71
RankFeatures	74.99 $\pm$ 12.18	19.80 $\pm$ 30.95	23.45 $\pm$ 33.12	60.99 $\pm$ 22.83
RSPool	76.66 $\pm$ 8.85	22.51 $\pm$ 15.23	27.45 $\pm$ 19.81	70.75 $\pm$ 15.18
SPECTRE	68.15 $\pm$ 7.69	11.19 $\pm$ 15.90	7.48 $\pm$ 16.54	52.02 $\pm$ 18.03
TIF	71.88 $\pm$ 9.91	13.87 $\pm$ 21.02	15.02 $\pm$ 25.81	55.52 $\pm$ 22.15
TopER	68.15 $\pm$ 7.69	12.70 $\pm$ 14.09	9.06 $\pm$ 15.94	57.28 $\pm$ 17.34
REToGCL	86.69 $\pm$ 3.32	43.54 $\pm$ 9.69	52.67 $\pm$ 10.50	85.32 $\pm$ 3.85
COLORS-3				
BalanceGCL	74.45 $\pm$ 2.46	73.22 $\pm$ 1.25	59.68 $\pm$ 2.02	73.21 $\pm$ 2.88
HistoGraph	46.83 $\pm$ 2.74	47.10 $\pm$ 3.14	34.95 $\pm$ 2.79	42.99 $\pm$ 2.36
UniImb	9.45 $\pm$ 1.43	0.91 $\pm$ 0.38	0.16 $\pm$ 0.19	4.00 $\pm$ 1.05
DiffLift	22.01 $\pm$ 0.16	17.75 $\pm$ 1.61	10.00 $\pm$ 0.96	16.34 $\pm$ 1.08
KhanGCL	53.49 $\pm$ 1.06	53.17 $\pm$ 1.24	36.44 $\pm$ 1.22	51.67 $\pm$ 1.05
MaxCutPool	85.97 $\pm$ 3.56	82.53 $\pm$ 3.93	75.95 $\pm$ 5.67	85.08 $\pm$ 3.61
SimplicialMP	82.00 $\pm$ 40.24	80.00 $\pm$ 44.72	80.00 $\pm$ 44.72	80.33 $\pm$ 43.98
DEL	9.76 $\pm$ 0.02	0.00 $\pm$ 0.00	0.00 $\pm$ 0.00	1.62 $\pm$ 0.00
CellCLAT	43.97 $\pm$ 0.76	41.29 $\pm$ 0.71	25.12 $\pm$ 0.91	42.80 $\pm$ 0.67
pLSTM	36.39 $\pm$ 0.94	32.61 $\pm$ 2.22	20.60 $\pm$ 0.41	31.79 $\pm$ 0.71
AbstainGNN	93.27 $\pm$ 1.06	90.51 $\pm$ 0.82	87.63 $\pm$ 1.90	92.66 $\pm$ 1.13
EdgePrompt	34.93 $\pm$ 0.54	31.55 $\pm$ 1.08	21.31 $\pm$ 0.41	30.09 $\pm$ 0.66
DualPrism	37.04 $\pm$ 1.26	34.80 $\pm$ 1.96	23.82 $\pm$ 0.51	32.41 $\pm$ 0.81
InvGNN	10.03 $\pm$ 0.01	0.00 $\pm$ 0.00	0.00 $\pm$ 0.00	1.66 $\pm$ 0.00
GenVsExp	54.23 $\pm$ 2.66	55.45 $\pm$ 2.68	39.50 $\pm$ 2.95	49.67 $\pm$ 3.21

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Table 6 — continued

Method	Accuracy	NMI	ARI	Macro-F1
KAGNN	38.06 ± 0.92	35.05 ± 2.07	23.93 ± 1.60	33.83 ± 1.70
MultiNet	37.07 ± 1.05	32.29 ± 1.90	21.90 ± 1.65	32.74 ± 1.61
HSPGKN	71.14 ± 8.75	68.36 ± 10.66	58.44 ± 12.30	68.04 ± 9.64
DGDA	33.67 ± 1.50	29.98 ± 1.89	20.31 ± 1.07	28.92 ± 1.82
RSNN	99.97 ± 0.04	99.93 ± 0.10	99.95 ± 0.08	99.97 ± 0.05
TopoFormer	28.53 ± 4.17	34.05 ± 2.84	15.16 ± 2.30	22.57 ± 3.89
G2PM	94.89 ± 5.55	93.31 ± 5.86	91.16 ± 9.07	94.25 ± 6.29
Git-TaskTree	75.26 ± 7.59	75.39 ± 10.61	65.54 ± 12.36	72.18 ± 8.05
GNN+	63.17 ± 3.04	64.75 ± 1.91	51.40 ± 3.64	59.57 ± 3.12
Hourglass	91.75 ± 4.66	90.39 ± 4.98	86.47 ± 7.40	90.62 ± 5.08
HyperEnc	92.34 ± 4.92	91.04 ± 4.29	87.37 ± 7.30	91.49 ± 5.53
LEAP	61.52 ± 4.12	56.38 ± 3.10	45.19 ± 3.14	57.98 ± 5.08
MagEdgePool	31.04 ± 3.63	26.63 ± 3.08	16.45 ± 3.07	26.13 ± 4.88
NodeID	50.09 ± 1.27	47.48 ± 0.74	34.53 ± 1.23	47.28 ± 1.58
PRUNE	32.89 ± 1.46	27.64 ± 2.22	16.84 ± 2.41	28.44 ± 1.69
RankFeatures	52.62 ± 5.45	46.91 ± 4.94	35.55 ± 5.37	49.37 ± 6.20
RSPool	46.15 ± 1.67	43.43 ± 1.29	31.63 ± 1.81	42.21 ± 1.75
SPECTRE	92.50 ± 7.21	91.52 ± 5.91	88.41 ± 9.10	91.50 ± 8.19
TIF	77.11 ± 3.30	80.05 ± 4.58	69.65 ± 5.76	72.61 ± 3.40
TopER	90.31 ± 8.01	90.09 ± 5.74	85.38 ± 9.46	88.93 ± 9.62
REToGCL	74.92 ± 1.59	70.70 ± 1.20	57.48 ± 2.03	74.20 ± 1.63
PTC_MR				
BalanceGCL	56.41 ± 6.71	2.04 ± 2.06	1.66 ± 3.15	55.67 ± 6.68
HistoGraph	56.40 ± 2.25	1.04 ± 1.44	0.45 ± 1.37	38.82 ± 7.07
UniImb	52.60 ± 4.91	0.11 ± 0.21	-0.33 ± 0.47	36.37 ± 3.78
DiffLift	52.29 ± 7.85	2.85 ± 6.36	0.83 ± 1.86	34.54 ± 2.85
KhanGCL	60.76 ± 3.79	3.44 ± 2.39	3.70 ± 3.25	60.07 ± 3.86
MaxCutPool	56.11 ± 0.70	0.90 ± 1.40	0.14 ± 0.36	40.54 ± 8.72
SimplicialMP	56.11 ± 0.70	0.17 ± 0.38	0.15 ± 0.34	37.49 ± 3.76
DEL	56.37 ± 3.09	1.00 ± 1.75	0.55 ± 1.94	43.38 ± 8.76
CellCLAT	56.46 ± 7.17	2.16 ± 1.29	1.89 ± 2.07	55.47 ± 7.21
pLSTM	56.11 ± 0.70	0.64 ± 1.43	0.16 ± 0.35	36.53 ± 1.61
AbstainGNN	57.86 ± 2.44	1.80 ± 1.21	1.29 ± 1.44	46.05 ± 7.96
EdgePrompt	58.70 ± 3.64	2.61 ± 2.94	2.09 ± 2.59	47.22 ± 7.22
DualPrism	54.96 ± 1.97	0.09 ± 0.18	-0.27 ± 0.60	39.39 ± 6.49
InvGNN	57.87 ± 3.87	1.75 ± 2.55	1.61 ± 3.20	41.87 ± 11.70
GenVsExp	56.99 ± 1.67	0.63 ± 0.83	0.67 ± 0.94	43.40 ± 8.41
KAGNN	56.97 ± 1.58	1.73 ± 2.64	0.67 ± 0.93	39.58 ± 4.07
MultiNet	56.70 ± 1.37	1.87 ± 2.69	0.50 ± 0.78	39.75 ± 4.09
HSPGKN	56.95 ± 3.67	1.87 ± 2.43	0.98 ± 2.52	47.15 ± 8.99
DGDA	52.04 ± 6.59	0.44 ± 0.72	0.18 ± 0.85	39.27 ± 7.42
RSNN	59.93 ± 5.20	4.39 ± 4.94	3.46 ± 4.44	51.09 ± 11.22
TopoFormer	56.40 ± 0.79	0.84 ± 1.35	0.27 ± 0.42	42.13 ± 7.13
G2PM	55.52 ± 0.63	0.00 ± 0.01	-0.13 ± 0.30	37.83 ± 4.50
Git-TaskTree	56.41 ± 3.41	1.24 ± 1.68	0.60 ± 2.10	51.27 ± 5.29
GNN+	55.29 ± 7.64	2.41 ± 1.79	1.60 ± 2.40	48.29 ± 11.46
Hourglass	54.64 ± 4.44	0.56 ± 1.09	0.10 ± 1.00	44.52 ± 9.53
HyperEnc	58.42 ± 3.47	2.27 ± 2.36	1.85 ± 2.51	48.83 ± 8.14
LEAP	56.97 ± 1.58	1.59 ± 2.61	0.67 ± 0.92	39.40 ± 5.09
MagEdgePool	57.85 ± 1.63	3.35 ± 2.20	1.20 ± 1.04	41.01 ± 4.46
NodeID	53.18 ± 5.60	0.49 ± 0.45	0.02 ± 0.84	42.09 ± 8.68
PRUNE	57.28 ± 2.57	2.44 ± 4.86	0.98 ± 1.82	40.22 ± 5.50
RankFeatures	57.55 ± 4.44	2.69 ± 3.32	1.52 ± 3.21	48.43 ± 6.31
RSPool	53.19 ± 8.65	1.58 ± 3.04	1.41 ± 4.42	47.65 ± 12.02
SPECTRE	54.30 ± 5.66	1.07 ± 1.72	0.35 ± 2.18	47.18 ± 7.00
TIF	55.21 ± 3.15	0.58 ± 1.13	0.02 ± 1.10	41.50 ± 4.54
TopER	58.13 ± 2.58	1.46 ± 1.76	1.48 ± 1.79	47.12 ± 9.10
REToGCL	63.39 ± 5.89	5.59 ± 4.09	6.93 ± 5.76	62.33 ± 6.10
Mutagenicity				
BalanceGCL	75.93 ± 1.60	20.02 ± 2.70	26.87 ± 3.40	75.54 ± 1.64
HistoGraph	79.89 ± 2.30	27.58 ± 4.59	35.82 ± 5.57	79.52 ± 2.36
UniImb	65.67 ± 3.82	7.39 ± 3.78	10.06 ± 4.95	63.77 ± 5.30
DiffLift	73.27 ± 2.62	16.20 ± 4.12	21.79 ± 5.07	72.97 ± 2.75
KhanGCL	77.56 ± 1.18	22.85 ± 2.16	30.34 ± 2.67	77.24 ± 1.22
MaxCutPool	77.38 ± 1.58	22.93 ± 3.01	29.97 ± 3.33	77.06 ± 1.57
SimplicialMP	74.54 ± 10.66	21.03 ± 11.88	27.43 ± 15.54	70.47 ± 19.07
DEL	80.68 ± 1.05	29.04 ± 2.10	37.60 ± 2.61	80.43 ± 1.07
CellCLAT	68.04 ± 2.34	9.26 ± 2.54	13.02 ± 3.33	66.72 ± 2.72
pLSTM	80.38 ± 1.63	28.40 ± 3.20	36.91 ± 3.89	80.10 ± 1.69
AbstainGNN	78.58 ± 2.32	25.01 ± 4.62	32.74 ± 5.40	78.16 ± 2.28
EdgePrompt	77.56 ± 1.20	23.18 ± 1.56	30.34 ± 2.61	77.21 ± 1.03
DualPrism	78.69 ± 1.85	25.17 ± 3.57	32.94 ± 4.25	78.22 ± 2.05
InvGNN	55.36 ± 0.01	0.00 ± 0.00	0.00 ± 0.01	35.73 ± 0.21
GenVsExp	79.78 ± 2.92	27.51 ± 6.17	35.64 ± 7.30	79.36 ± 2.95
KAGNN	78.76 ± 3.51	25.54 ± 6.64	33.39 ± 8.07	78.34 ± 3.79
MultiNet	76.39 ± 1.15	21.30 ± 1.95	27.80 ± 2.40	76.05 ± 1.17
HSPGKN	77.61 ± 2.49	23.33 ± 4.13	30.57 ± 5.52	76.96 ± 2.94
DGDA	77.15 ± 2.74	22.38 ± 4.84	29.62 ± 6.07	76.65 ± 2.87
RSNN	78.17 ± 1.16	23.97 ± 2.22	31.68 ± 2.67	77.86 ± 1.23
TopoFormer	74.91 ± 1.29	18.58 ± 2.18	24.78 ± 2.57	74.61 ± 1.42
G2PM	77.77 ± 2.47	23.91 ± 3.80	30.92 ± 5.42	77.08 ± 3.07
Git-TaskTree	79.73 ± 1.73	26.95 ± 3.46	35.36 ± 4.11	79.37 ± 1.81
GNN+	79.69 ± 0.83	27.33 ± 1.49	35.19 ± 1.97	79.40 ± 0.78
Hourglass	79.34 ± 1.39	27.00 ± 3.08	34.41 ± 3.24	79.08 ± 1.56
HyperEnc	79.62 ± 2.98	27.28 ± 5.83	35.28 ± 7.04	79.36 ± 3.08
LEAP	78.99 ± 2.60	26.08 ± 4.68	33.75 ± 5.84	78.64 ± 2.98
MagEdgePool	78.19 ± 1.11	24.18 ± 2.01	31.72 ± 2.48	77.76 ± 0.99
NodeID	72.45 ± 1.65	15.19 ± 1.51	20.14 ± 2.90	72.19 ± 1.54
PRUNE	77.54 ± 1.81	23.09 ± 3.19	30.33 ± 4.05	76.99 ± 1.87
RankFeatures	77.56 ± 1.38	23.40 ± 2.72	30.36 ± 3.06	77.21 ± 1.40
RSPool	79.59 ± 2.24	26.80 ± 4.39	35.10 ± 5.30	79.27 ± 2.34
SPECTRE	78.02 ± 2.57	24.21 ± 4.21	31.52 ± 5.70	77.50 ± 3.06
TIF	79.25 ± 0.76	26.19 ± 1.68	34.14 ± 1.77	78.80 ± 0.64
TopER	78.81 ± 2.39	25.35 ± 4.31	33.29 ± 5.35	78.40 ± 2.46
REToGCL	77.33 ± 1.65	22.47 ± 3.09	29.88 ± 3.69	76.98 ± 1.79
AIDS				
BalanceGCL	99.45 ± 0.45	94.20 ± 4.37	97.49 ± 2.04	99.14 ± 0.70

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Table 6 — continued

Method	Accuracy	NMI	ARI	Macro-F1
HistoGraph	93.50 ± 1.08	56.47 ± 5.40	71.97 ± 3.92	89.77 ± 1.46
UniImb	96.85 ± 2.97	77.68 ± 14.87	85.51 ± 13.92	94.52 ± 5.76
DiffLift	79.75 ± 0.43	0.05 ± 0.09	-0.15 ± 0.20	44.60 ± 0.38
KhanGCL	99.50 ± 0.31	94.59 ± 2.66	97.70 ± 1.41	99.21 ± 0.49
MaxCutPool	99.30 ± 0.27	92.81 ± 2.45	96.79 ± 1.26	98.89 ± 0.44
SimplicialMP	99.65 ± 0.42	96.41 ± 4.00	98.39 ± 1.92	99.45 ± 0.67
DEL	97.15 ± 0.96	75.77 ± 6.51	87.22 ± 4.22	95.50 ± 1.53
CellCLAT	99.30 ± 0.21	92.93 ± 1.71	96.78 ± 0.96	98.89 ± 0.34
pLSTM	93.75 ± 1.50	57.28 ± 7.27	73.34 ± 5.89	90.38 ± 2.27
AbstainGNN	98.90 ± 1.07	90.15 ± 8.83	94.96 ± 4.87	98.23 ± 1.73
EdgePrompt	82.30 ± 5.14	9.66 ± 21.60	12.19 ± 27.26	52.43 ± 17.86
DualPrism	91.65 ± 2.00	48.64 ± 7.87	65.48 ± 7.18	87.34 ± 2.81
InvGNN	98.90 ± 0.74	90.14 ± 5.46	94.94 ± 3.41	98.23 ± 1.22
GenVsExp	95.55 ± 1.47	66.75 ± 7.72	80.60 ± 6.00	93.11 ± 2.18
KAGNN	92.70 ± 2.46	52.76 ± 11.90	69.17 ± 9.89	88.65 ± 4.00
MultiNet	86.65 ± 5.05	27.23 ± 19.28	39.93 ± 25.34	72.42 ± 16.64
HSPGKN	96.60 ± 0.80	72.62 ± 5.12	84.71 ± 3.63	94.55 ± 1.36
DGDA	92.85 ± 1.58	52.26 ± 7.09	68.95 ± 6.77	88.41 ± 2.91
RSNN	99.30 ± 0.60	93.37 ± 5.18	96.78 ± 2.75	98.88 ± 0.96
TopoFormer	94.40 ± 1.01	61.03 ± 5.02	76.06 ± 4.04	91.48 ± 1.51
G2PM	98.95 ± 0.78	89.67 ± 7.40	95.23 ± 3.53	98.35 ± 1.23
Git-TaskTree	95.25 ± 2.26	65.28 ± 13.07	79.18 ± 9.65	92.45 ± 3.70
GNN+	98.05 ± 1.11	82.97 ± 6.37	91.11 ± 5.13	96.86 ± 1.92
Hourglass	99.35 ± 0.52	93.32 ± 5.06	97.02 ± 2.38	98.97 ± 0.82
HyperEnc	99.15 ± 0.60	91.77 ± 5.38	96.10 ± 2.77	98.65 ± 0.97
LEAP	93.00 ± 2.80	53.99 ± 13.63	69.66 ± 11.87	88.58 ± 4.91
MagEdgePool	83.55 ± 4.90	13.87 ± 19.12	20.94 ± 28.73	59.08 ± 20.05
NodeID	98.45 ± 0.60	85.74 ± 4.71	93.02 ± 2.62	97.60 ± 0.91
PRUNE	80.00 ± 0.00	0.00 ± 0.00	0.00 ± 0.00	44.44 ± 0.00
RankFeatures	92.90 ± 2.97	54.18 ± 13.57	68.31 ± 13.75	87.68 ± 6.45
RSPool	96.10 ± 1.76	70.34 ± 10.53	82.74 ± 7.25	93.82 ± 2.66
SPECTRE	99.40 ± 0.34	93.51 ± 3.17	97.25 ± 1.54	99.05 ± 0.53
TIF	93.15 ± 3.44	55.71 ± 16.32	70.93 ± 13.89	89.19 ± 5.65
TopER	99.45 ± 0.41	94.31 ± 4.01	97.48 ± 1.89	99.13 ± 0.65
ReToGCL	99.40 ± 0.28	93.69 ± 2.40	97.24 ± 1.31	99.05 ± 0.46
BZR				
BalanceGCL	82.94 ± 6.90	21.18 ± 20.42	32.91 ± 23.02	70.32 ± 12.42
HistoGraph	78.77 ± 0.42	0.00 ± 0.00	0.00 ± 0.00	44.06 ± 0.13
UniImb	67.16 ± 25.82	0.00 ± 0.00	0.00 ± 0.00	38.70 ± 11.94
DiffLift	78.77 ± 0.42	0.00 ± 0.00	0.00 ± 0.00	44.06 ± 0.13
KhanGCL	83.93 ± 3.01	21.61 ± 6.80	37.36 ± 8.01	74.37 ± 3.68
MaxCutPool	78.77 ± 0.42	0.00 ± 0.00	0.00 ± 0.00	44.06 ± 0.13
SimplicialMP	78.77 ± 0.42	0.00 ± 0.00	0.00 ± 0.00	44.06 ± 0.13
DEL	78.77 ± 0.42	0.00 ± 0.00	0.00 ± 0.00	44.06 ± 0.13
CellCLAT	80.73 ± 4.47	12.15 ± 11.89	20.05 ± 15.95	61.97 ± 10.46
pLSTM	78.77 ± 0.42	0.00 ± 0.00	0.00 ± 0.00	44.06 ± 0.13
AbstainGNN	78.77 ± 0.42	0.00 ± 0.00	0.00 ± 0.00	44.06 ± 0.13
EdgePrompt	78.77 ± 0.42	0.00 ± 0.00	0.00 ± 0.00	44.06 ± 0.13
DualPrism	82.97 ± 5.63	16.64 ± 20.65	25.20 ± 25.54	63.26 ± 16.16
InvGNN	78.77 ± 0.42	0.00 ± 0.00	0.00 ± 0.00	44.06 ± 0.13
GenVsExp	81.48 ± 6.22	11.37 ± 25.42	14.01 ± 31.32	53.17 ± 20.41
KAGNN	78.77 ± 0.42	0.00 ± 0.00	0.00 ± 0.00	44.06 ± 0.13
MultiNet	78.77 ± 0.42	0.00 ± 0.00	0.00 ± 0.00	44.06 ± 0.13
HSPGKN	78.77 ± 0.42	0.00 ± 0.00	0.00 ± 0.00	44.06 ± 0.13
DGDA	77.06 ± 4.22	0.03 ± 0.06	-0.49 ± 1.09	44.82 ± 1.57
RSNN	78.77 ± 0.42	0.00 ± 0.00	0.00 ± 0.00	44.06 ± 0.13
TopoFormer	78.77 ± 0.42	0.00 ± 0.00	0.00 ± 0.00	44.06 ± 0.13
G2PM	78.77 ± 0.42	0.00 ± 0.00	0.00 ± 0.00	44.06 ± 0.13
Git-TaskTree	78.77 ± 0.42	0.00 ± 0.00	0.00 ± 0.00	44.06 ± 0.13
GNN+	78.77 ± 0.42	0.00 ± 0.00	0.00 ± 0.00	44.06 ± 0.13
Hourglass	78.77 ± 0.42	0.00 ± 0.00	0.00 ± 0.00	44.06 ± 0.13
HyperEnc	79.27 ± 1.18	2.52 ± 5.63	2.61 ± 5.83	46.29 ± 4.99
LEAP	79.00 ± 2.70	3.60 ± 7.84	7.09 ± 14.23	50.97 ± 11.84
MagEdgePool	78.77 ± 0.42	0.00 ± 0.00	0.00 ± 0.00	44.06 ± 0.13
NodeID	80.00 ± 1.51	6.11 ± 5.71	12.95 ± 12.01	55.98 ± 11.08
PRUNE	78.77 ± 0.42	0.00 ± 0.00	0.00 ± 0.00	44.06 ± 0.13
RankFeatures	78.04 ± 2.04	1.21 ± 2.72	3.13 ± 7.00	47.96 ± 8.60
RSPool	79.51 ± 1.36	3.87 ± 5.70	3.96 ± 5.89	47.46 ± 5.08
SPECTRE	78.77 ± 0.42	0.00 ± 0.00	0.00 ± 0.00	44.06 ± 0.13
TIF	66.92 ± 25.68	0.09 ± 0.21	0.53 ± 1.18	39.61 ± 12.61
TopER	78.77 ± 0.42	0.00 ± 0.00	0.00 ± 0.00	44.06 ± 0.13
ReToGCL	83.93 ± 6.42	24.22 ± 18.52	37.03 ± 19.92	73.10 ± 9.65
COX2				
BalanceGCL	83.08 ± 1.46	18.23 ± 4.68	31.66 ± 5.09	70.48 ± 3.16
HistoGraph	78.16 ± 0.46	0.00 ± 0.00	0.00 ± 0.00	43.87 ± 0.14
UniImb	78.16 ± 0.46	0.00 ± 0.00	0.00 ± 0.00	43.87 ± 0.14
DiffLift	78.16 ± 0.46	0.00 ± 0.00	0.00 ± 0.00	43.87 ± 0.14
KhanGCL	82.66 ± 2.63	17.43 ± 6.56	31.84 ± 8.17	70.94 ± 4.55
MaxCutPool	78.16 ± 0.46	0.00 ± 0.00	0.00 ± 0.00	43.87 ± 0.14
SimplicialMP	78.16 ± 0.46	0.00 ± 0.00	0.00 ± 0.00	43.87 ± 0.14
DEL	78.16 ± 0.46	0.00 ± 0.00	0.00 ± 0.00	43.87 ± 0.14
CellCLAT	76.88 ± 1.51	0.80 ± 1.02	2.93 ± 3.79	48.75 ± 4.08
pLSTM	78.16 ± 0.46	0.00 ± 0.00	0.00 ± 0.00	43.87 ± 0.14
AbstainGNN	78.16 ± 0.46	0.00 ± 0.00	0.00 ± 0.00	43.87 ± 0.14
EdgePrompt	78.38 ± 0.79	1.43 ± 2.50	1.88 ± 2.65	45.77 ± 2.71
DualPrism	78.16 ± 0.46	0.00 ± 0.00	0.00 ± 0.00	43.87 ± 0.14
InvGNN	78.16 ± 0.46	0.00 ± 0.00	0.00 ± 0.00	43.87 ± 0.14
GenVsExp	78.81 ± 4.55	5.91 ± 11.56	7.06 ± 18.80	49.77 ± 14.03
KAGNN	78.16 ± 0.46	0.00 ± 0.00	0.00 ± 0.00	43.87 ± 0.14
MultiNet	78.16 ± 0.46	0.00 ± 0.00	0.00 ± 0.00	43.87 ± 0.14
HSPGKN	78.16 ± 0.46	0.00 ± 0.00	0.00 ± 0.00	43.87 ± 0.14
DGDA	68.05 ± 22.42	0.10 ± 0.23	0.76 ± 1.71	40.61 ± 7.24
RSNN	78.16 ± 0.46	0.00 ± 0.00	0.00 ± 0.00	43.87 ± 0.14
TopoFormer	78.16 ± 0.46	0.00 ± 0.00	0.00 ± 0.00	43.87 ± 0.14
G2PM	78.16 ± 0.46	0.00 ± 0.00	0.00 ± 0.00	43.87 ± 0.14
Git-TaskTree	67.19 ± 25.57	2.17 ± 4.86	2.21 ± 4.94	40.54 ± 13.46
GNN+	78.16 ± 0.46	0.00 ± 0.00	0.00 ± 0.00	43.87 ± 0.14
Hourglass	78.37 ± 0.41	1.67 ± 3.74	3.98 ± 8.89	47.95 ± 9.03

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Table 6 — continued

Method	Accuracy	NMI	ARI	Macro-F1
HyperEnc	77.09 ± 2.26	0.64 ± 1.42	2.16 ± 4.84	46.96 ± 6.98
LEAP	78.38 ± 0.79	2.04 ± 4.56	4.65 ± 10.39	48.39 ± 10.16
MagEdgePool	78.16 ± 0.46	0.00 ± 0.00	0.00 ± 0.00	43.87 ± 0.14
NodeID	77.31 ± 1.33	0.58 ± 0.80	2.57 ± 3.53	47.71 ± 5.23
PRUNE	78.16 ± 0.46	0.00 ± 0.00	0.00 ± 0.00	43.87 ± 0.14
RankFeatures	78.16 ± 0.46	0.00 ± 0.00	0.00 ± 0.00	43.87 ± 0.14
RSPool	78.16 ± 0.46	0.00 ± 0.00	0.00 ± 0.00	43.87 ± 0.14
SPECTRE	78.16 ± 0.46	1.99 ± 4.44	4.54 ± 10.15	48.46 ± 10.32
TIF	67.10 ± 25.02	0.00 ± 0.00	0.00 ± 0.00	38.78 ± 11.47
TopER	78.59 ± 1.22	1.70 ± 3.80	2.84 ± 6.34	46.46 ± 5.86
ReToGCL	82.01 ± 2.12	16.67 ± 5.57	29.99 ± 4.00	70.08 ± 1.30
ClinTox				
BalanceGCL	92.23 ± 0.89	6.71 ± 3.48	14.42 ± 5.15	38.99 ± 2.27
HistoGraph	92.43 ± 0.27	0.00 ± 0.00	0.00 ± 0.00	32.02 ± 0.05
UniImb	92.43 ± 0.27	0.00 ± 0.00	0.00 ± 0.00	32.02 ± 0.05
DiffLift	92.43 ± 0.27	0.00 ± 0.00	0.00 ± 0.00	32.02 ± 0.05
KhanGCL	92.57 ± 0.93	7.36 ± 5.80	12.67 ± 5.89	38.03 ± 2.42
MaxCutPool	92.43 ± 0.27	0.00 ± 0.00	0.00 ± 0.00	32.02 ± 0.05
SimplicialMP	92.43 ± 0.27	0.00 ± 0.00	0.00 ± 0.00	32.02 ± 0.05
DEL	92.57 ± 0.32	2.24 ± 5.00	3.78 ± 8.45	33.78 ± 3.92
CellCLAT	91.89 ± 0.44	1.16 ± 0.89	2.77 ± 3.39	33.74 ± 1.67
pLSTM	92.43 ± 0.27	0.00 ± 0.00	0.00 ± 0.00	32.02 ± 0.05
AbstainGNN	92.43 ± 0.27	0.00 ± 0.00	0.00 ± 0.00	32.02 ± 0.05
EdgePrompt	92.43 ± 0.27	0.00 ± 0.00	0.00 ± 0.00	32.02 ± 0.05
DualPrism	92.43 ± 0.27	0.00 ± 0.00	0.00 ± 0.00	32.02 ± 0.05
InvGNN	92.43 ± 0.27	0.00 ± 0.00	0.00 ± 0.00	32.02 ± 0.05
GenVsExp	92.37 ± 0.35	0.03 ± 0.07	-0.12 ± 0.27	32.01 ± 0.06
KAGNN	92.43 ± 0.27	0.00 ± 0.00	0.00 ± 0.00	32.02 ± 0.05
MultiNet	92.43 ± 0.27	0.00 ± 0.00	0.00 ± 0.00	32.02 ± 0.05
HSPGKN	92.43 ± 0.27	0.00 ± 0.00	0.00 ± 0.00	32.02 ± 0.05
DGDA	92.37 ± 0.35	0.03 ± 0.07	-0.12 ± 0.27	32.01 ± 0.06
RSNN	92.43 ± 0.27	0.00 ± 0.00	0.00 ± 0.00	32.02 ± 0.05
TopoFormer	92.43 ± 0.27	0.00 ± 0.00	0.00 ± 0.00	32.02 ± 0.05
G2PM	92.43 ± 0.27	0.00 ± 0.00	0.00 ± 0.00	32.02 ± 0.05
Git-TaskTree	92.43 ± 0.27	0.00 ± 0.00	0.00 ± 0.00	32.02 ± 0.05
GNN+	92.43 ± 0.27	0.00 ± 0.00	0.00 ± 0.00	32.02 ± 0.05
Hourglass	92.43 ± 0.27	0.00 ± 0.00	0.00 ± 0.00	32.02 ± 0.05
HyperEnc	92.43 ± 0.27	0.00 ± 0.00	0.00 ± 0.00	32.02 ± 0.05
LEAP	92.43 ± 0.27	0.00 ± 0.00	0.00 ± 0.00	32.02 ± 0.05
MagEdgePool	92.43 ± 0.27	0.00 ± 0.00	0.00 ± 0.00	32.02 ± 0.05
NodeID	92.43 ± 0.27	0.00 ± 0.00	0.00 ± 0.00	32.02 ± 0.05
PRUNE	92.43 ± 0.27	0.00 ± 0.00	0.00 ± 0.00	32.02 ± 0.05
RankFeatures	92.43 ± 0.27	0.00 ± 0.00	0.00 ± 0.00	32.02 ± 0.05
RSPool	92.43 ± 0.27	0.00 ± 0.00	0.00 ± 0.00	32.02 ± 0.05
SPECTRE	92.43 ± 0.27	0.00 ± 0.00	0.00 ± 0.00	32.02 ± 0.05
TIF	92.43 ± 0.27	0.00 ± 0.00	0.00 ± 0.00	32.02 ± 0.05
TopER	92.43 ± 0.27	0.00 ± 0.00	0.00 ± 0.00	32.02 ± 0.05
ReToGCL	92.43 ± 0.95	9.52 ± 5.69	17.75 ± 8.89	40.42 ± 3.62
BACE				
BalanceGCL	76.01 ± 1.94	20.38 ± 3.30	26.94 ± 4.12	75.82 ± 1.93
HistoGraph	65.82 ± 6.45	8.24 ± 5.33	11.01 ± 7.08	64.20 ± 7.30
UniImb	53.08 ± 1.93	0.94 ± 1.34	-0.19 ± 0.28	37.71 ± 5.69
DiffLift	59.41 ± 5.55	3.73 ± 3.53	4.13 ± 4.30	56.18 ± 9.40
KhanGCL	76.41 ± 3.00	21.19 ± 5.32	27.94 ± 6.54	76.12 ± 3.02
MaxCutPool	63.58 ± 1.85	5.34 ± 1.79	7.15 ± 2.04	62.19 ± 1.21
SimplicialMP	59.42 ± 4.71	3.08 ± 2.98	3.77 ± 3.48	51.53 ± 14.94
DEL	75.35 ± 0.66	19.92 ± 1.04	25.46 ± 1.33	75.14 ± 0.53
CellCLAT	70.99 ± 1.08	12.85 ± 1.43	17.37 ± 1.79	70.44 ± 1.09
pLSTM	54.86 ± 1.17	0.22 ± 0.50	0.32 ± 0.71	39.33 ± 9.23
AbstainGNN	60.20 ± 3.94	4.35 ± 2.63	4.30 ± 3.15	59.18 ± 5.03
EdgePrompt	71.11 ± 7.34	14.60 ± 7.94	19.28 ± 10.26	70.80 ± 7.70
DualPrism	66.56 ± 3.97	8.88 ± 4.46	11.16 ± 5.42	65.67 ± 3.32
InvGNN	56.77 ± 3.65	1.23 ± 1.91	1.64 ± 2.70	43.98 ± 12.10
GenVsExp	65.90 ± 4.64	10.20 ± 7.94	10.43 ± 6.90	63.68 ± 5.61
KAGNN	54.33 ± 0.07	0.00 ± 0.00	0.00 ± 0.00	35.20 ± 0.03
MultiNet	54.33 ± 0.07	0.00 ± 0.00	0.00 ± 0.00	35.20 ± 0.03
HSPGKN	65.83 ± 5.57	9.19 ± 4.76	10.67 ± 6.83	64.05 ± 8.08
DGDA	65.69 ± 6.08	9.39 ± 5.41	10.70 ± 7.74	64.70 ± 6.00
RSNN	64.06 ± 7.35	7.00 ± 6.01	9.22 ± 7.89	59.00 ± 14.66
TopoFormer	54.86 ± 1.57	0.40 ± 0.89	0.40 ± 0.89	41.57 ± 9.84
G2PM	61.73 ± 5.65	5.16 ± 3.26	6.12 ± 4.85	58.83 ± 6.81
Git-TaskTree	66.04 ± 3.38	8.32 ± 3.26	10.28 ± 4.09	63.42 ± 5.57
GNN+	66.28 ± 7.55	10.57 ± 7.79	12.07 ± 11.15	64.31 ± 9.30
Hourglass	65.89 ± 3.90	7.84 ± 4.00	10.26 ± 4.99	64.74 ± 5.17
HyperEnc	65.57 ± 3.67	7.82 ± 3.69	9.78 ± 4.96	63.47 ± 5.09
LEAP	68.27 ± 6.74	11.68 ± 7.53	14.48 ± 9.43	66.88 ± 7.57
MagEdgePool	54.46 ± 1.42	0.43 ± 0.93	0.16 ± 0.54	37.80 ± 3.56
NodeID	70.26 ± 3.10	12.77 ± 4.31	16.43 ± 5.20	69.36 ± 3.97
PRUNE	56.37 ± 4.53	1.43 ± 3.20	1.61 ± 3.61	41.05 ± 13.06
RankFeatures	66.68 ± 7.15	10.00 ± 6.60	12.46 ± 9.45	66.39 ± 7.25
RSPool	68.33 ± 5.18	10.65 ± 5.58	14.00 ± 7.12	67.64 ± 6.27
SPECTRE	60.68 ± 9.27	5.86 ± 9.73	6.87 ± 12.16	55.30 ± 14.16
TIF	64.25 ± 4.96	7.63 ± 4.55	8.49 ± 6.88	60.37 ± 7.53
TopER	63.97 ± 7.06	8.07 ± 7.41	8.98 ± 8.05	59.02 ± 13.71
ReToGCL	77.46 ± 1.50	22.89 ± 2.85	30.01 ± 3.37	77.31 ± 1.55
BBBP				
BalanceGCL	86.02 ± 1.40	29.35 ± 4.85	44.79 ± 5.26	77.99 ± 2.73
HistoGraph	83.96 ± 1.41	23.69 ± 4.21	36.50 ± 4.40	73.23 ± 2.63
UniImb	77.79 ± 2.94	4.14 ± 9.25	6.12 ± 13.68	48.47 ± 11.48
DiffLift	77.54 ± 1.21	3.85 ± 4.10	5.41 ± 6.50	48.60 ± 6.22
KhanGCL	86.37 ± 1.21	30.53 ± 4.01	45.48 ± 4.75	78.16 ± 2.52
MaxCutPool	82.83 ± 3.12	20.60 ± 10.22	30.89 ± 11.54	69.07 ± 7.13
SimplicialMP	84.01 ± 2.93	23.38 ± 8.39	36.14 ± 13.68	72.31 ± 9.87
DEL	86.41 ± 2.28	31.92 ± 7.21	47.14 ± 7.79	79.39 ± 3.80
CellCLAT	83.13 ± 1.51	20.36 ± 4.57	34.19 ± 6.03	72.13 ± 3.82
pLSTM	84.41 ± 4.56	24.68 ± 14.21	38.05 ± 21.61	72.60 ± 16.47
AbstainGNN	83.28 ± 4.11	21.93 ± 13.32	32.15 ± 18.45	68.87 ± 14.41

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Table 6 — continued

Method	Accuracy	NMI	ARI	Macro-F1
EdgePrompt	82.54 ± 3.41	18.78 ± 10.51	28.89 ± 16.26	66.93 ± 13.26
DualPrism	83.47 ± 4.00	21.71 ± 12.46	32.44 ± 18.38	68.94 ± 14.40
InvGNN	76.51 ± 0.08	0.00 ± 0.00	0.00 ± 0.00	43.35 ± 0.03
GenVsExp	84.06 ± 1.69	23.43 ± 5.03	37.56 ± 6.23	74.04 ± 3.66
KAGNN	83.38 ± 3.87	21.05 ± 11.80	33.77 ± 18.90	70.26 ± 15.06
MultiNet	77.88 ± 2.38	5.20 ± 7.15	10.58 ± 12.61	53.79 ± 11.21
HSPGKN	82.49 ± 2.44	18.94 ± 7.01	29.42 ± 11.30	68.08 ± 8.29
DGDA	83.52 ± 2.82	24.29 ± 5.50	38.25 ± 7.92	75.15 ± 4.50
RSNN	85.73 ± 1.18	28.39 ± 3.79	43.54 ± 4.36	77.30 ± 2.28
TopoFormer	82.98 ± 1.90	22.09 ± 3.61	36.69 ± 4.13	74.58 ± 2.08
G2PM	83.28 ± 2.69	22.02 ± 8.34	34.00 ± 10.85	71.47 ± 7.19
Git-TaskTree	85.43 ± 0.76	27.74 ± 3.01	41.78 ± 2.84	76.16 ± 1.62
GNN+	82.20 ± 3.96	18.16 ± 11.99	28.84 ± 18.99	67.00 ± 14.86
Hourglass	84.75 ± 1.65	26.28 ± 5.46	40.52 ± 6.80	75.81 ± 4.11
HyperEnc	81.22 ± 2.85	15.75 ± 9.30	27.27 ± 15.56	67.01 ± 13.40
LEAP	82.30 ± 3.42	18.09 ± 10.34	28.17 ± 15.90	66.59 ± 12.72
MagEdgePool	78.67 ± 3.05	7.19 ± 9.00	13.16 ± 15.37	55.43 ± 13.22
NodeID	84.11 ± 1.12	23.83 ± 2.82	38.20 ± 3.81	74.61 ± 2.45
PRUNE	81.31 ± 2.83	15.22 ± 8.76	26.47 ± 14.88	66.19 ± 12.80
RankFeatures	83.27 ± 3.99	21.54 ± 12.79	32.31 ± 19.67	68.99 ± 15.26
RSPool	85.68 ± 1.48	28.89 ± 4.55	43.92 ± 5.71	77.65 ± 3.37
SPECTRE	83.33 ± 2.26	21.74 ± 6.01	33.62 ± 10.72	71.09 ± 7.38
TIF	84.36 ± 2.15	25.98 ± 4.39	40.35 ± 6.08	76.12 ± 3.48
TopER	86.12 ± 0.89	30.00 ± 3.11	45.15 ± 3.66	78.20 ± 2.20
REToGCL	86.61 ± 1.61	31.43 ± 5.37	47.20 ± 5.86	79.28 ± 2.98
SIDER				
BalanceGCL	75.51 ± 11.37	1.33 ± 1.70	2.13 ± 3.11	49.24 ± 5.19
HistoGraph	75.43 ± 11.74	0.95 ± 1.63	1.40 ± 2.71	47.32 ± 5.54
UniImb	73.22 ± 15.22	0.11 ± 0.43	0.12 ± 1.08	43.12 ± 5.50
DiffLift	75.04 ± 12.38	0.48 ± 1.11	0.42 ± 1.48	45.32 ± 4.44
KhanGCL	75.47 ± 11.40	1.34 ± 1.78	2.35 ± 3.32	49.49 ± 5.11
MaxCutPool	75.31 ± 11.97	0.50 ± 1.15	0.54 ± 1.34	45.43 ± 4.59
SimplicialMP	75.21 ± 12.17	0.60 ± 1.69	0.66 ± 2.00	44.88 ± 4.22
DEL	75.39 ± 11.61	1.29 ± 1.91	2.31 ± 3.61	48.74 ± 5.49
CellCLAT	74.75 ± 11.64	1.31 ± 1.68	2.44 ± 4.10	49.92 ± 4.56
pLSTM	75.39 ± 11.88	0.43 ± 1.05	0.59 ± 1.48	45.21 ± 5.07
AbstainGNN	75.09 ± 12.25	0.52 ± 1.11	0.55 ± 1.51	45.88 ± 5.08
EdgePrompt	75.63 ± 11.62	0.70 ± 1.54	1.00 ± 2.09	46.58 ± 5.22
DualPrism	75.43 ± 11.83	0.77 ± 1.60	1.26 ± 2.75	47.27 ± 5.77
InvGNN	74.92 ± 12.59	0.57 ± 1.68	0.68 ± 2.06	44.56 ± 4.49
GenVsExp	74.88 ± 11.95	0.77 ± 1.16	1.29 ± 2.56	48.13 ± 5.04
KAGNN	75.33 ± 12.00	0.50 ± 1.12	0.62 ± 1.44	45.61 ± 4.86
MultiNet	75.25 ± 12.09	0.41 ± 0.98	0.38 ± 1.02	45.00 ± 4.16
HSPGKN	75.34 ± 11.92	0.93 ± 1.60	1.57 ± 2.58	47.93 ± 4.96
DGDA	75.42 ± 11.79	1.16 ± 1.95	2.07 ± 3.42	48.19 ± 5.99
RSNN	75.32 ± 12.03	0.67 ± 1.44	0.93 ± 2.08	45.47 ± 4.14
TopoFormer	75.43 ± 11.75	0.59 ± 1.41	0.80 ± 1.95	45.84 ± 4.92
G2PM	75.47 ± 11.77	0.95 ± 1.67	1.67 ± 2.86	47.86 ± 5.36
Git-TaskTree	75.45 ± 11.76	0.99 ± 1.74	1.66 ± 2.98	47.72 ± 5.82
GNN+	75.78 ± 11.48	1.00 ± 2.23	1.52 ± 3.13	47.29 ± 6.01
Hourglass	75.62 ± 11.54	1.00 ± 1.73	1.59 ± 2.77	47.83 ± 5.85
HyperEnc	75.54 ± 11.71	0.93 ± 1.69	1.39 ± 2.34	47.21 ± 5.27
LEAP	75.62 ± 11.52	1.13 ± 1.93	2.02 ± 3.22	48.26 ± 5.52
MagEdgePool	75.27 ± 12.11	0.48 ± 1.32	0.48 ± 1.37	45.03 ± 4.58
NodeID	75.43 ± 11.74	1.25 ± 2.26	2.22 ± 4.19	48.71 ± 5.68
PRUNE	75.26 ± 12.11	0.37 ± 1.08	0.42 ± 1.29	44.74 ± 4.59
RankFeatures	75.56 ± 11.62	1.14 ± 1.80	1.86 ± 2.99	47.98 ± 5.47
RSPool	75.52 ± 11.72	0.86 ± 1.68	1.30 ± 2.45	47.30 ± 5.32
SPECTRE	75.58 ± 11.66	0.95 ± 1.74	1.52 ± 2.78	47.38 ± 5.89
TIF	75.59 ± 11.63	1.20 ± 1.97	2.15 ± 3.23	48.79 ± 5.85
TopER	75.51 ± 11.69	0.93 ± 1.63	1.41 ± 2.54	47.58 ± 5.54
REToGCL	74.92 ± 11.53	1.42 ± 1.55	3.03 ± 3.71	51.04 ± 4.72
HIV				
BalanceGCL	96.72 ± 0.08	11.08 ± 2.09	18.55 ± 2.51	59.17 ± 1.32
HistoGraph	96.52 ± 0.06	5.11 ± 3.11	9.95 ± 5.97	54.58 ± 3.28
UniImb	96.50 ± 0.02	1.26 ± 2.83	2.43 ± 5.44	50.45 ± 2.99
DiffLift	96.49 ± 0.10	3.47 ± 3.87	6.87 ± 7.62	52.90 ± 4.20
KhanGCL	96.69 ± 0.04	9.78 ± 0.89	16.49 ± 1.09	58.07 ± 0.58
MaxCutPool	96.58 ± 0.08	5.18 ± 3.17	8.69 ± 5.30	53.85 ± 2.89
SimplicialMP	96.53 ± 0.07	2.46 ± 2.98	4.05 ± 4.59	51.32 ± 2.49
DEL	96.70 ± 0.06	12.24 ± 4.05	21.06 ± 8.41	60.53 ± 4.53
CellCLAT	96.59 ± 0.04	5.53 ± 1.07	9.29 ± 1.63	54.19 ± 0.88
pLSTM	96.79 ± 0.28	13.40 ± 12.31	22.38 ± 20.56	61.12 ± 11.03
AbstainGNN	96.60 ± 0.10	8.16 ± 3.89	14.55 ± 7.39	57.05 ± 4.03
EdgePrompt	96.56 ± 0.05	7.46 ± 2.41	13.82 ± 5.73	56.68 ± 3.15
DualPrism	96.49 ± 0.01	0.00 ± 0.00	0.00 ± 0.00	49.11 ± 0.00
InvGNN	96.54 ± 0.08	2.38 ± 3.46	3.81 ± 5.30	51.18 ± 2.87
GenVsExp	96.72 ± 0.16	11.22 ± 4.81	18.72 ± 7.19	59.26 ± 3.83
KAGNN	96.49 ± 0.01	0.00 ± 0.00	0.00 ± 0.00	49.11 ± 0.00
MultiNet	96.49 ± 0.01	0.00 ± 0.00	0.00 ± 0.00	49.11 ± 0.00
HSPGKN	96.53 ± 0.05	3.23 ± 4.43	5.99 ± 8.23	52.39 ± 4.51
DGDA	96.67 ± 0.23	11.17 ± 8.79	19.74 ± 14.69	59.80 ± 7.89
RSNN	96.93 ± 0.10	18.88 ± 3.02	31.06 ± 4.96	65.82 ± 2.63
TopoFormer	96.49 ± 0.01	0.00 ± 0.00	0.00 ± 0.00	49.11 ± 0.00
G2PM	96.69 ± 0.09	11.64 ± 2.52	20.17 ± 5.54	60.06 ± 3.02
Git-TaskTree	96.63 ± 0.19	7.50 ± 8.76	12.94 ± 15.16	56.11 ± 8.18
GNN+	96.86 ± 0.16	18.51 ± 4.25	31.80 ± 6.29	66.25 ± 3.30
Hourglass	96.70 ± 0.06	12.53 ± 4.49	22.26 ± 8.68	61.19 ± 4.69
HyperEnc	96.74 ± 0.13	12.75 ± 5.94	21.81 ± 10.34	60.91 ± 5.54
LEAP	96.73 ± 0.19	13.86 ± 4.19	24.03 ± 6.47	62.13 ± 3.44
MagEdgePool	96.49 ± 0.01	0.00 ± 0.00	0.00 ± 0.00	49.11 ± 0.00
NodeID	96.61 ± 0.09	10.94 ± 2.62	20.37 ± 4.80	60.22 ± 2.60
PRUNE	96.49 ± 0.01	0.00 ± 0.00	0.00 ± 0.00	49.11 ± 0.00
RankFeatures	96.66 ± 0.12	9.76 ± 5.86	16.96 ± 10.65	58.32 ± 5.79
RSPool	96.49 ± 0.01	0.00 ± 0.00	0.00 ± 0.00	49.11 ± 0.00
SPECTRE	96.62 ± 0.16	10.01 ± 8.28	18.30 ± 14.70	59.05 ± 7.93
TIF	96.78 ± 0.09	13.35 ± 4.07	22.27 ± 7.37	61.15 ± 3.96
TopER	96.77 ± 0.02	15.35 ± 3.01	26.79 ± 6.42	63.60 ± 3.46

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Table 6 — continued

Method	Accuracy	NMI	ARI	Macro-F1
REToGCL	96.62 ± 0.08	8.76 ± 1.46	15.60 ± 1.73	57.63 ± 0.91
Tox21				
BalanceGCL	92.64 ± 4.40	9.41 ± 11.20	16.88 ± 16.70	58.38 ± 9.06
HistoGraph	92.65 ± 4.65	9.39 ± 13.33	15.59 ± 20.24	57.22 ± 11.42
UniImb	91.92 ± 5.36	1.15 ± 3.62	2.65 ± 7.77	49.87 ± 4.93
DiffLift	92.25 ± 4.76	1.68 ± 4.77	3.40 ± 9.15	50.10 ± 5.50
KhanGCL	92.63 ± 4.42	9.50 ± 11.26	17.25 ± 16.70	58.63 ± 9.08
MaxCutPool	92.38 ± 4.80	4.63 ± 10.70	7.54 ± 16.92	52.15 ± 9.61
SimplicialMP	92.58 ± 4.51	4.98 ± 10.53	8.42 ± 16.93	53.03 ± 9.86
DEL	92.99 ± 4.28	11.85 ± 13.53	19.25 ± 20.45	59.57 ± 11.45
CellCLAT	92.13 ± 4.77	2.09 ± 2.65	4.74 ± 5.33	51.34 ± 3.11
pLSTM	92.73 ± 4.48	7.83 ± 13.39	12.56 ± 20.91	55.25 ± 12.08
AbstainGNN	92.46 ± 4.59	4.04 ± 9.40	7.03 ± 15.69	52.19 ± 9.14
EdgePrompt	92.90 ± 4.26	9.10 ± 12.44	15.13 ± 19.75	57.09 ± 11.27
DualPrism	92.66 ± 4.51	8.67 ± 12.19	14.87 ± 19.21	56.97 ± 10.80
InvGNN	92.25 ± 4.72	0.18 ± 1.01	0.42 ± 2.34	48.25 ± 1.95
GenVsExp	92.77 ± 4.49	10.22 ± 12.73	17.43 ± 19.52	58.59 ± 10.90
KAGNN	92.50 ± 4.66	5.78 ± 11.09	9.67 ± 18.12	53.49 ± 10.50
MultiNet	92.34 ± 4.83	3.99 ± 10.07	6.41 ± 15.72	51.42 ± 8.93
HSPGKN	92.38 ± 4.89	6.07 ± 12.05	10.05 ± 18.38	53.85 ± 10.39
DGDA	92.54 ± 4.58	7.14 ± 11.29	11.81 ± 18.25	54.90 ± 10.42
RSNN	92.67 ± 4.56	7.99 ± 12.33	12.94 ± 19.11	55.53 ± 11.09
TopoFormer	92.43 ± 4.76	6.21 ± 10.70	10.95 ± 17.75	54.40 ± 10.27
G2PM	92.57 ± 4.65	7.32 ± 11.82	12.45 ± 18.72	55.43 ± 10.63
Git-TaskTree	92.63 ± 4.53	7.56 ± 11.60	13.04 ± 18.65	55.86 ± 10.52
GNN+	92.87 ± 4.42	9.69 ± 14.40	15.43 ± 21.56	57.10 ± 12.27
Hourglass	92.79 ± 4.46	9.31 ± 13.06	15.48 ± 20.34	57.19 ± 11.59
HyperEnc	92.69 ± 4.57	8.58 ± 12.85	14.10 ± 20.18	56.32 ± 11.51
LEAP	92.63 ± 4.66	8.30 ± 12.78	13.50 ± 19.26	55.97 ± 10.76
MagEdgePool	92.35 ± 4.80	3.45 ± 9.56	5.56 ± 15.16	50.99 ± 8.59
NodeID	92.58 ± 4.66	7.61 ± 11.49	12.73 ± 17.44	55.55 ± 9.78
PRUNE	92.32 ± 4.77	3.08 ± 8.44	5.22 ± 14.17	50.82 ± 8.15
RankFeatures	92.75 ± 4.38	8.32 ± 11.96	13.92 ± 18.68	56.30 ± 10.52
RSPool	92.69 ± 4.63	8.21 ± 13.90	12.89 ± 20.89	55.42 ± 11.78
SPECTRE	92.68 ± 4.55	7.93 ± 11.53	13.37 ± 18.15	55.97 ± 10.39
TIF	92.65 ± 4.61	8.13 ± 11.89	13.70 ± 18.53	56.19 ± 10.41
TopER	92.69 ± 4.65	9.07 ± 13.55	14.86 ± 20.73	56.71 ± 11.76
REToGCL	92.71 ± 4.37	11.44 ± 11.87	20.79 ± 16.62	60.70 ± 8.90
ABIDE				
BalanceGCL	51.56 ± 3.06	0.28 ± 0.22	-0.09 ± 0.35	50.76 ± 3.42
HistoGraph	54.47 ± 3.15	1.02 ± 0.93	0.72 ± 0.96	49.58 ± 1.29
UniImb	52.66 ± 3.23	0.45 ± 0.56	0.19 ± 0.82	48.07 ± 8.74
DiffLift	51.77 ± 4.87	0.89 ± 1.44	0.47 ± 1.15	47.47 ± 4.35
KhanGCL	52.06 ± 2.53	0.24 ± 0.45	-0.12 ± 0.68	51.36 ± 2.71
MaxCutPool	52.76 ± 1.42	0.33 ± 0.45	0.12 ± 0.32	37.76 ± 7.81
SimplicialMP	52.06 ± 0.27	0.23 ± 0.39	-0.08 ± 0.14	37.46 ± 7.13
DEL	49.66 ± 5.36	0.92 ± 0.56	0.49 ± 0.63	46.43 ± 5.37
CellCLAT	53.45 ± 2.65	0.45 ± 0.60	0.25 ± 0.84	50.88 ± 3.09
pLSTM	51.57 ± 4.32	0.44 ± 0.52	0.36 ± 0.51	42.16 ± 8.80
AbstainGNN	52.96 ± 2.50	0.74 ± 1.15	0.32 ± 0.83	37.41 ± 6.78
EdgePrompt	51.36 ± 2.79	0.22 ± 0.29	-0.06 ± 0.26	44.78 ± 4.70
DualPrism	50.56 ± 3.45	0.27 ± 0.32	-0.10 ± 0.42	50.10 ± 3.54
InvGNN	52.96 ± 1.58	0.46 ± 1.03	0.21 ± 0.47	36.40 ± 4.65
GenVsExp	52.96 ± 3.42	0.91 ± 1.15	0.37 ± 0.93	45.52 ± 4.09
KAGNN	51.96 ± 0.67	0.04 ± 0.09	-0.05 ± 0.12	35.14 ± 1.84
MultiNet	52.26 ± 0.01	0.00 ± 0.00	0.00 ± 0.00	34.32 ± 0.00
HSPGKN	54.38 ± 4.71	1.16 ± 1.16	1.01 ± 1.37	52.60 ± 3.71
DGDA	54.18 ± 4.66	1.99 ± 1.48	1.05 ± 0.86	46.29 ± 2.90
RSNN	53.06 ± 1.54	0.37 ± 0.67	0.22 ± 0.44	37.27 ± 5.51
TopoFormer	54.16 ± 1.90	0.71 ± 0.92	0.50 ± 0.72	43.61 ± 6.04
G2PM	53.58 ± 4.54	1.60 ± 1.70	0.82 ± 0.99	45.83 ± 4.13
Git-TaskTree	51.97 ± 5.38	0.94 ± 0.91	0.72 ± 0.73	44.56 ± 5.51
GNN+	51.66 ± 1.96	0.28 ± 0.52	-0.09 ± 0.38	41.59 ± 6.13
Hourglass	53.37 ± 3.86	1.15 ± 1.31	0.56 ± 1.00	46.74 ± 6.40
HyperEnc	53.98 ± 4.42	1.75 ± 1.32	0.90 ± 0.68	46.69 ± 1.79
LEAP	52.27 ± 5.04	0.98 ± 1.03	0.62 ± 0.91	47.03 ± 4.05
MagEdgePool	50.76 ± 3.35	0.17 ± 0.39	0.15 ± 0.32	35.25 ± 2.08
NodeID	52.27 ± 4.74	1.01 ± 1.14	0.55 ± 0.93	46.01 ± 3.59
PRUNE	52.26 ± 0.01	0.00 ± 0.00	0.00 ± 0.00	34.32 ± 0.00
RankFeatures	55.27 ± 2.28	1.32 ± 1.25	0.94 ± 0.94	47.16 ± 4.14
RSPool	50.76 ± 3.23	0.23 ± 0.34	-0.04 ± 0.38	45.14 ± 6.01
SPECTRE	52.37 ± 5.47	1.44 ± 1.49	0.81 ± 1.01	46.04 ± 4.85
TIF	51.97 ± 4.63	1.04 ± 1.55	0.51 ± 0.67	42.45 ± 5.43
TopER	52.47 ± 4.48	1.16 ± 1.20	0.53 ± 0.53	45.05 ± 4.78
REToGCL	54.37 ± 1.90	0.56 ± 0.57	0.41 ± 0.76	52.51 ± 1.80
ADHD-200				
BalanceGCL	62.75 ± 6.04	3.78 ± 5.64	6.28 ± 8.20	58.61 ± 7.56
HistoGraph	61.83 ± 0.00	0.00 ± 0.00	0.00 ± 0.00	38.21 ± 0.00
UniImb	56.34 ± 10.29	0.10 ± 0.23	-0.32 ± 0.71	36.70 ± 5.24
DiffLift	61.53 ± 0.68	0.01 ± 0.03	-0.10 ± 0.23	38.76 ± 1.23
KhanGCL	67.02 ± 4.50	6.71 ± 4.85	10.98 ± 6.71	63.66 ± 4.70
MaxCutPool	61.83 ± 0.00	0.00 ± 0.00	0.00 ± 0.00	38.21 ± 0.00
SimplicialMP	61.68 ± 0.34	0.01 ± 0.03	0.06 ± 0.13	39.13 ± 2.06
DEL	59.85 ± 4.02	0.87 ± 1.82	-0.20 ± 0.98	42.36 ± 6.08
CellCLAT	60.46 ± 3.56	1.39 ± 1.74	2.90 ± 3.20	54.31 ± 3.69
pLSTM	61.83 ± 0.00	0.00 ± 0.00	0.00 ± 0.00	38.21 ± 0.00
AbstainGNN	61.83 ± 0.00	0.00 ± 0.00	0.00 ± 0.00	38.21 ± 0.00
EdgePrompt	62.75 ± 2.61	1.68 ± 2.43	2.73 ± 3.56	47.24 ± 8.94
DualPrism	63.36 ± 1.95	1.96 ± 2.16	3.65 ± 3.74	49.48 ± 10.66
InvGNN	61.83 ± 0.00	0.00 ± 0.00	0.00 ± 0.00	38.21 ± 0.00
GenVsExp	64.27 ± 2.93	3.83 ± 4.96	5.21 ± 5.84	51.98 ± 14.02
KAGNN	61.83 ± 0.00	0.00 ± 0.00	0.00 ± 0.00	38.21 ± 0.00
MultiNet	61.83 ± 0.00	0.00 ± 0.00	0.00 ± 0.00	38.21 ± 0.00
HSPGKN	62.75 ± 3.72	2.91 ± 2.35	4.21 ± 4.69	53.00 ± 9.24
DGDA	62.29 ± 4.50	2.58 ± 2.69	4.09 ± 4.55	53.18 ± 8.82
RSNN	61.83 ± 0.00	0.00 ± 0.00	0.00 ± 0.00	38.21 ± 0.00
TopoFormer	61.83 ± 0.00	0.00 ± 0.00	0.00 ± 0.00	38.21 ± 0.00

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Table 6 — continued

Method	Accuracy	NMI	ARI	Macro-F1
G2PM	63.05 ± 2.89	3.34 ± 2.05	3.65 ± 3.30	50.23 ± 9.95
Git-TaskTree	61.68 ± 3.83	1.82 ± 2.28	2.54 ± 4.09	49.03 ± 10.94
GNN+	61.83 ± 0.00	0.00 ± 0.00	0.00 ± 0.00	38.21 ± 0.00
Hourglass	61.98 ± 2.05	1.58 ± 2.26	2.57 ± 3.64	49.05 ± 11.65
HyperEnc	63.66 ± 2.57	2.22 ± 3.10	3.70 ± 4.99	47.94 ± 12.21
LEAP	64.27 ± 7.21	5.52 ± 4.24	8.64 ± 6.92	61.44 ± 5.69
MagEdgePool	61.83 ± 0.00	0.00 ± 0.00	0.00 ± 0.00	38.21 ± 0.00
NodeID	61.68 ± 4.56	1.97 ± 3.08	3.26 ± 5.12	51.04 ± 9.94
PRUNE	61.83 ± 0.00	0.00 ± 0.00	0.00 ± 0.00	38.21 ± 0.00
RankFeatures	63.66 ± 2.89	3.37 ± 2.53	5.09 ± 4.10	54.30 ± 8.27
RSPool	61.22 ± 1.37	0.08 ± 0.19	0.25 ± 0.56	41.10 ± 6.48
SPECTRE	63.82 ± 3.76	3.51 ± 3.47	4.64 ± 5.12	50.89 ± 10.29
TIF	61.22 ± 5.38	2.34 ± 1.49	2.70 ± 4.09	49.76 ± 10.15
TopER	62.75 ± 2.61	1.56 ± 2.45	1.98 ± 4.22	43.51 ± 9.21
REToGCL	66.26 ± 4.23	5.88 ± 4.06	9.90 ± 5.83	62.86 ± 4.55